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Research Report

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Artificial Intelligence and Machine Learning for Space Domain Awareness

Characterizing the Impact on Mission Effectiveness



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About This Report

The U.S. Space Force (USSF) space domain awareness (SDA) mission is complex and rapidly growing in complexity. Space objects and activities have differing priorities for observation, and the SDA mission needs to optimize collection and analysis according to these priorities. For example, high-interest objects may require a higher revisit rate for sensor collection than an object characterized as space debris. In addition to these varying priorities, the sheer data volume, the complexities of orbital mechanics and space operations, and the time constraints introduced in an operational warfighting domain, present significant challenges to those tasked with the SDA mission and point to this mission as a prime candidate for support from artificial intelligence and machine learning (AI/ML) tools.

The work should be of interest to USSF leadership—in particular, the Office of the Assistant Secretary for Space Acquisition and Integration and Space Systems Command—and to USSF personnel tasked with carrying out the SDA mission, notably those at the 18th and 19th Space Defense Squadrons.

This work is accompanied by a companion report that describes the technical details of two AI/ML tools developed in support of this research, *Artificial Intelligence and Machine Learning for Space Domain Awareness: The Development of Two Artificial Intelligence Case Studies* by Jonathan Tran, Prateek Puri, Jordan Logue, Anthony Jacques, Li Ang Zhang, Krista Langeland, George Nacouzi, and Gary J. Briggs (RR-A2318-2).

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Summary

Issue

To address the growing demands of operating in the space domain, space domain awareness (SDA) operators must determine how to prioritize sensor observations more effectively, scale up to meet the sheer volume of resident space objects, and develop analytic capabilities that reflect the complexity of orbital mechanics and space operations, all while maintaining the responsiveness necessitated by operations in a warfighting domain. These factors present significant challenges to those tasked with the SDA mission and point to this mission as a prime candidate for support from artificial intelligence (AI) and machine learning (ML) tools, because such tools have the potential to increase the analysis tempo, expand the amount of usable data for this analysis, and free up operator time for more-complex tasks. AI/ML tools can help SDA operators meet these increasing challenges. For example, leveraging AI/ML to optimize sensor scheduling could improve the accuracy of the space catalog by allowing more-efficient use of existing sensing resources. AI/ML tools can also facilitate the calculation of close proximity risks and provide indications and warning by flagging abnormal events.

Although AI/ML tools have the potential to help meet these SDA challenges, the impact of these tools on the overall success of the SDA mission is not well understood, and this lack of understanding is a barrier to planning and optimizing the integration of the tools. This research seeks to develop insight into the following research questions:

- How can existing AI/ML and automation technologies benefit the SDA mission?
- Which processes within the SDA mission area can be improved by increased automation and AI/ML?
- What are the prerequisites and enabling technologies that are necessary for AI/ML implementation?
- What limitations should be considered when employing these technologies?

Approach

Our approach for this research focused on characterizing the nature of the impact that AI/ML tools could bring to the SDA mission. To accomplish this, we examined the processes that make up this mission, the metrics for these processes, the opportunities for impact, and the impact of these tools on the process and the outcomes. The research presented here focuses in detail on the conjunction assessment (CA) mission to enable us to look at a singular SDA process in depth. CA is the probabilistic process by which the 18th and 19th Space Defense Squadrons (SDSs) quantify the risk of collision in space.

To support this assessment, we interviewed stakeholders, reviewed existing academic and doctrinal literature, developed detailed process maps, and built exploratory AI/ML models.

Key Findings

- Because of the growing demands and changing nature of the SDA mission, AI/ML tools have a high opportunity for impact if they can be force multipliers for SDA operators.
- AI/ML tools cannot address all the challenges of the SDA mission, but process changes can help these tools achieve greater impact.
- Realizing significant impacts from AI/ML requires moving to an architecture that enables more AI/ML development and fielding.
- AI/ML tool development that supports more-optimized sensor tasking could have a cascading impact on the rest of the SDA mission.
- Better quantification of risk and uncertainty tolerance can support the improved performance of AI/ML tools focused on prediction and classification.

Recommendations

- Space Systems Command SDA Acquisition Delta (SSC/SZG) and the Air Force Research Laboratory should provide clear guidance to AI/ML tool developers to focus on tools that address the needs of operators today but could also benefit a future SDA architecture.
- AI/ML tool development should focus on force multipliers that help meet the growing detection and characterization challenges faced by the 18th and 19th SDSs. SSC/SZG should look for opportunities to invest in these tools.
- SDA operators—in particular, at the 18th SDS, 19th SDS, and National Space Defense Center—should seek ways to articulate mission needs, sensor requirements, and acceptable uncertainty to optimize potential enhancements from AI/ML tools.
- Space Operations Command, via the SDA Mission Area Team, should examine where SDA processes could be modified to enable more-significant impact from AI/ML tools.
- AI/ML tool development should focus on those tools that enable more-effective and more-efficient sensor tasking, and SSC/SZG should seek ways to acquire and develop those tools. Efforts at the 18th SDS to capture observation intent should also be supported, as these efforts, in combination, are key enablers.
- SSC/SZG should continue to support efforts in AI/ML tool development and ensure that there are processes and infrastructure in place to test and validate these models.
- The availability of high-quality training data is another enabler for AI/ML impact, and USSF should support efforts at Space Systems Command in its Cross Mission Data team to ensure the availability of these data to AI/ML tool developers.

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Chapter 1

Introduction: The Changing Space Domain Awareness Mission

The U.S. Department of Defense (DoD) acknowledges the critical role of space in all-domain operations in future conflict and recognizes that understanding the role of space technologies is critical for optimizing investment in and operationalization of these technologies in support of all-domain operations. Gen James Dickinson, while commander of U.S. Space Command (USSPACECOM), named space domain awareness (SDA) as the command's number one need.[1]

Artificial intelligence (AI) is a broad field with competing definitions from various academic fields; for the purposes of this report, we consider it alongside machine learning (ML) as providers of the following technological capabilities: computer vision, language processing, reinforcement learning, prediction and classification, generative learning, and expert systems.[2] We focus on understanding how these capabilities can help the U.S. Space Force (USSF) in the SDA mission.

AI/ML has the potential to establish superior SDA and enable rapid decisionmaking in the SDA domain, outpacing that of adversaries. Understanding how to optimally use these technologies and plan for their expected impact on the SDA mission will help improve mission effectiveness, optimize resource management, and inform investment decisions. However, the potential benefits from the application of AI/ML to the SDA mission is not well understood. In particular, the U.S. Air Force and USSF would like to know what the impact of such tools might be and, for those high-impact areas, how they can invest to support the implementation of these tools.

Study Purpose

To build this understanding, we developed a framework for identifying how existing AI/ML technologies could be applied to support decisionmaking in support of SDA and to assess the material benefits of such implementations. Specifically, we had the following research questions:

- How can existing AI/ML and automation technologies benefit the SDA mission?
- Which processes within the SDA mission area can be improved by increased automation and AI/ML?

[1] Erwin, "Space Domain Awareness."

[2] We drew from U.S. Code, Title 15, Section 9401, which provides policy-oriented definitions for AI and ML. *Artificial intelligence* is defined as "a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments." *Machine learning* is "an application of artificial intelligence that is characterized by providing systems the ability to automatically learn and improve on the basis of data or experience."

- What are the prerequisites and enabling technologies that are necessary for AI/ML implementation?
- What limitations should be considered when employing these technologies?

Methodology: Change Impact Analysis

To characterize the potential impact of AI/ML tools on the SDA mission, we examined the detailed processes that make up this mission and the metrics for these processes, identified opportunities for impact through AI/ML tools, and explored the potential change these tools would have on the process and the outcomes. The analysis focuses in detail on the conjunction assessment (CA) mission to enable in-depth examination of a singular SDA process. CA is foundational to other SDA processes, as it includes sensor tasking, data ingestion, characterization of movement, and correlation to known objects. All of these processes support other SDA processes, such as threat characterization and launch detection. We can therefore look at CA as both a case study and a foundational element within SDA more broadly.

To first assess the impact of change from AI/ML tools, the RAND team conducted interviews with stakeholders and executed a review of existing academic and doctrinal literature to build a view of what the SDA mission is, how it is changing, and what that could mean for AI/ML tool implementation and impact. The team conducted in-person interviews with operators from each cell within the 18th and 19th Space Defense Squadrons (SDSs), including both uniformed and contractor personnel. The team also engaged with the Kobayashi Maru software factory to gain insight into the infrastructure modernization efforts to support the SDA mission and potential AI applications.[3] To understand other SDA efforts, the team conducted virtual and in-person interviews with various organizations within the Space Systems Command (SSC) to learn about acquisitions and existing programs, including the AI and Autonomy Sandbox. **In total, we interviewed 23 stakeholders across nine organizations to gain a broad understanding of the mission.**

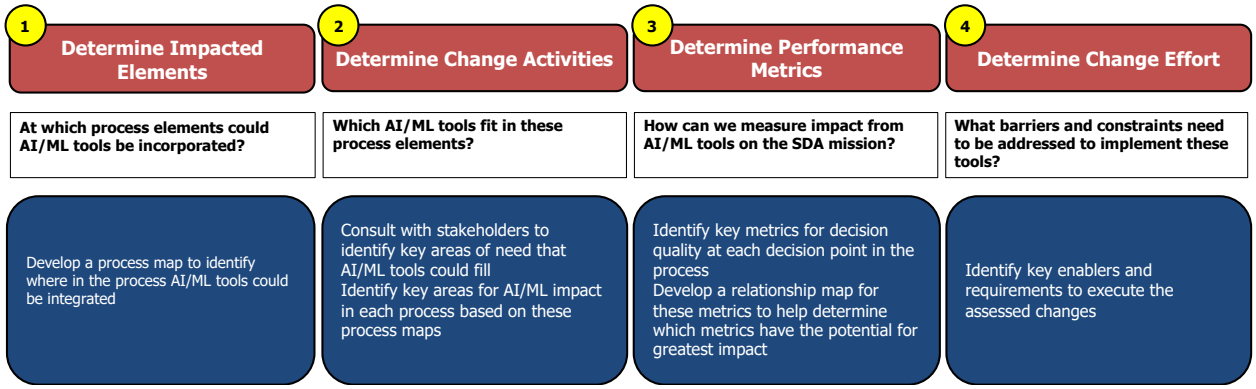
Recognizing the critical importance of understanding the intricate details of the SDA mission and operator needs, we had a researcher to spend a week with the 18th SDS. Spending time with operators allowed for a more in-depth understanding of the unique challenges in the daily SDA operations. We then used this foundation to inform our impact assessment.

To ensure that our impact assessment was systematic, we followed steps in the change-impact-analysis methodology used in the development of change management processes to better understand the holistic impact of deliberate changes on a system.[4] This methodology consists of four steps, described in Figure 1.1.

[3] See Chapter 2 for background about the Kobayashi Maru program.

[4] Rebentisch et al., "Assessment of Changes in Technical Systems and Their Effects on Cost and Duration Based on Structural Complexity."

Figure 1.1. Steps in Change-Impact-Analysis Methodology



SOURCE: Features information from Dietzmann and Alt, "Assessing the Business Impact of Artificial Intelligence."

The first step, *determine impacted elements*, aims to determine at which points AI/ML tools may be incorporated into a process. To identify these points, we engaged with the available literature and worked with stakeholders to develop a process map for a specific SDA mission: CA. We engaged with stakeholders at the operational level to identify where their needs are most acute and therefore capture where there is high potential for impact from AI/ML tools. This research is presented in Chapter 2.

The next step, *determine change activities*, looks at AI/ML tools to assess which tools would be appropriate for the different process points identified above. To accomplish this step, we surveyed AI/ML tools more broadly and provided an overview of which tools may be best suited for the CA process. This research is presented in conjunction with the first step in Chapter 2.

The third step, *determine performance metrics*, requires identifying metrics for the overall process and submetrics for each point at which AI/ML tools may be applied. We identified these metrics and used them to develop models to help understand the impact of two specific types of improvements that AI/ML could bring to the overall CA process. These models are discussed in Chapter 3.

The last step, *determine change effort*, seeks to capture the enablers and barriers to implementing AI/ML tools in this process. Although this report does not specifically focus on how to address these enablers and barriers, Chapter 4 discusses what they are and how addressing them could change the nature of the impact of AI/ML tools on the CA process.

The change-impact-analysis framework enabled us to identify key areas in which AI/ML could improve and support the CA process and facilitated the characterization of the impact from these tools. We then used this output to inform the design and building of two proof-of-concept tools to explore their application and potential for implementation in more detail.

Chapter 2

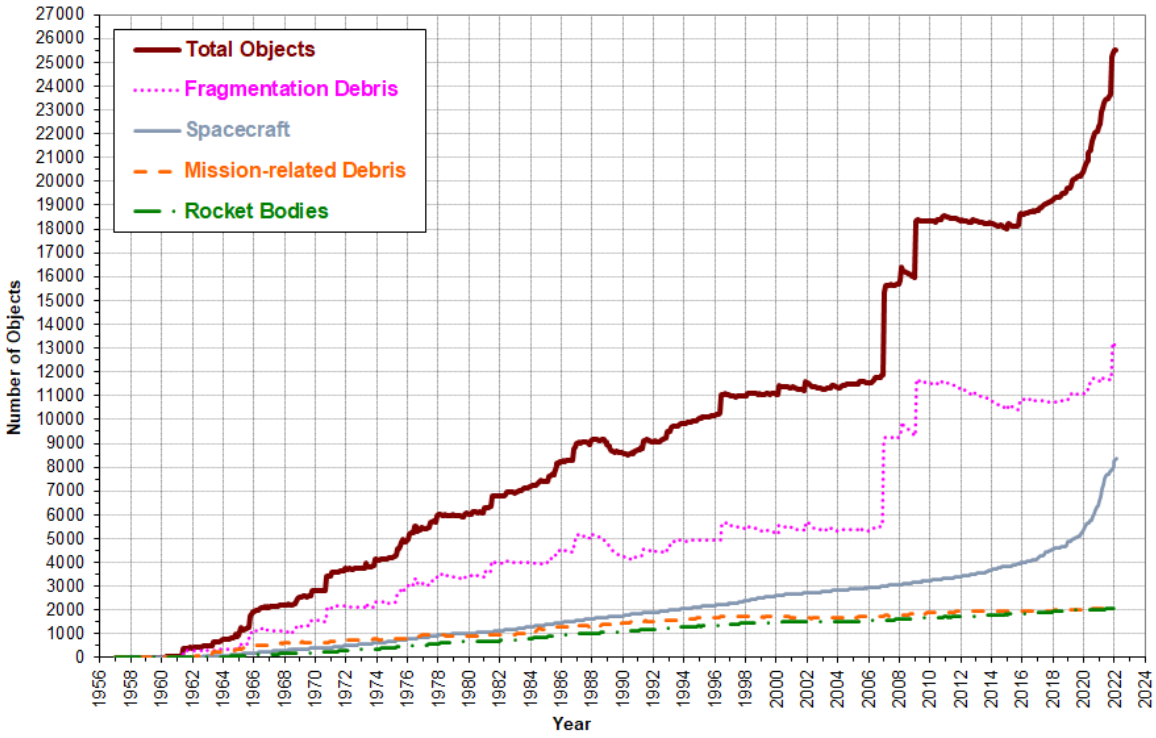
Background: The Space Domain Awareness Mission

SDA is vital to maintaining safe and secure access to space, deterring adversary actions in space, and achieving space superiority should a crisis or conflict occur. Gen B. Chance Saltzman, chief of space operations for the USSF, identified "comprehensive and actionable" awareness of the space domain as a core tenet of his theory of success.[5] However, SDA is increasingly challenged by the launch of new constellations, maneuverable assets, and activities from noncooperative actors in space. The U.S. Space Surveillance Network (SSN) currently tracks about 9,000 active payloads, 17,000 analyst objects, and 19,000 pieces of debris, for a total of about 45,000 objects overall, and maintains these in a catalog.[6] That number is growing dramatically from launches, collisions, and testing of direct-ascent antisatellite weapons, as shown in Figure 2.1.

[5] Hadley, "Saltzman Unveils 'Competitive Endurance' Theory to Guide Space Force."

[6] Space-Track.org, homepage.

Figure 2.1. Tracked Space Objects, as of April 2022



SOURCE: Reprinted from Astromaterials Research and Exploration Science, National Aeronautics and Space Administration, "Legend."

To meet this challenge of sustaining and improving SDA, the USSF needs to incorporate new tools to better leverage SSN resources in a way that can scale with the growth of activity and complexity in the SDA mission.[7] Using tools to address this change and growth are crucial for maintaining safety in space and ensuring the ability to protect and defend U.S. assets in orbit. Quicker response times are necessary, and improvements in SDA decisionmaking timelines are essential for achieving USSF goals for tactically responsive space capabilities.[8]

Managing congestion by tracking and forecasting space object trajectories is also crucial for improving awareness in support of space defense. The increasing amount of data generated by new and maneuverable space objects will require more analytic capabilities.[9] Effective defense relies on improved ability to characterize and forecast as strategic competitors increase their presence and level

[7] The emerging challenges facing the SDA mission are twofold: growing congestion and increasing competition. See, for example, Brock, "Space Delta 2." As the number of satellite constellations increases and strategic competition grows in intensity, SDA is moving beyond tracking existing space assets and debris to managing congestion and increasing focus on space defense. As a result, decisionmaking needs in support of SDA are changing and growing in complexity.

[8] Marrow, "Space Force Eyes 2025–2026 Timeframe for Tactically Responsive Space Capabilities."

[9] SDA stakeholders recognize this need. For example, Gen Stephen Whiting, former head of Space Operations Command and currently the new commander for USSPACECOM, acknowledged the potential for AI/ML to affect the SDA mission, noting that "we have a lot of data and AI/ML can help us parse through that data" (quoted in Wolfe, "U.S. Space Force Looks to AI/ML to Aid Space Domain Awareness").

of activity in the space domain.[10] The requirements for improving SDA provide an outline for how AI/ML technologies can play a critical role in establishing superior SDA and enabling better and faster decisionmaking than adversaries.

SDA is a priority mission area for the United States, and we have noted the changing SDA mission needs, but the United States also currently lacks strategic understanding of the operational advantages that AI/ML could provide to the SDA mission. Addressing this shortfall and determining how to leverage AI/ML in the SDA mission requires an in-depth understanding of the benefits and efficiencies that these tools could provide, what data are needed to support these tools, and which technologies may be available in the near term to support SDA. The output of this work will support the U.S. Department of the Air Force's decisionmaking as it assesses opportunities for using AI/ML in SDA and will provide benefit to the broader space community, including the National Reconnaissance Office, the National Aeronautics and Space Administration, the National Geospatial-Intelligence Agency, the Air Force Research Laboratory, Defense Advanced Research Projects Agency, and the Space Data Association.

Defining the SDA Mission: What Are the Objectives?

SDA is one of the five core competencies of the USSF and "encompasses the effective identification, characterization and understanding of any factor associated with the space domain that could affect space operations and thereby impact the security, safety, economy, or environment of our Nation."[11] Although the concept of SDA is not new,[12] the mission has evolved from maintaining catalogs and assessing conjunction threats of benign objects with limited maneuverability to addressing a threat environment that now includes a greater number of highly maneuverable and potentially noncooperative objects.[13]

Data for SDA originate from various sources, including the SSN, commercial assets, and allies and partners that contribute data from both ground- and space-based sensors. Contributing sensors

[10] As noted by one former space control officer with the former U.S. Air Force Space Command, the ability to characterize and forecast objects in space is crucial for effective defense: "The United States can't stop China from moving its own satellites. 'But what I certainly want to do is know when they're moving around and be able to forecast a little bit to know if I need to defend myself or protect myself.' . . . 'And so that is really the crux of it'" (Erwin, "Space Domain Awareness," quoting Brian Young).

[11] USSF, *Spacepower*.

[12] Space situational awareness (SSA) has been around since the beginning of the space era, when the U.S. government initiated a space surveillance program for detecting, tracking, and cataloging space objects. See, for example, Weeden and Cefola, "Computer Systems and Algorithms for Space Situational Awareness"; Stares, *The Militarization of Space*. In October 2019, then–Air Force Space Command redesignated SSA as SDA to highlight the shift of space from a benign environment to a warfighting domain due to emerging space and terrestrial threats. This shift was deemed necessary because, historically, SSA focused on elements that include surveillance (astrodynamics environmental data), as well as performing such activities as data collection, catalog, CA, and support for space traffic management functions, with most of these outputs captured in a publicly available catalog (Space-Track.org, homepage). SSA was redesignated SDA to reflect the growing acknowledgment of space as a warfighting domain, "beyond the Space Situational Awareness mindset of a benign environment to achieve a more effective and comprehensive SDA" (Erwin, "Air Force"). Recent doctrinal guidance (Joint Publication 3-14, *Space Operations*) identifies SSA as a subset of SDA that focuses on the orbital segment in particular to learn about and characterize space objects for space safety and sustainability.

[13] These objects include space debris, orbital assets in need of servicing, or assets controlled by noncooperative owner-operators (O/Os).

are dedicated SDA sensors and nondedicated sensors that support early warning and missile detection.

SDA Functions: What Does SDA Do?

SSA is a subset of SDA that focuses on the orbital segment. The functions of SDA and SSA are outlined in the January 2022 space doctrine note from the USSF and the 2020 Joint Publication 3-14.[14] These functions include detect/track/identify (D/T/ID), characterization, threat warning and assessment, and data integration and exploitation. Descriptions of these functions are provided in Table 2.1. As the SDA mission becomes more complex, these functions are becoming increasingly important and resource intensive. Exploiting data quickly and with high confidence is critical. Each of these SDA functions has key enablers that can improve the performance of this function. These enablers are noted in Table 2.1.

Table 2.1. Key Enablers for Improved SDA Mission Effectiveness

SDA Function	Description	Key Enablers
D/T/ID	Observation-based function that searches, discovers, and tracks spacecraft and events; categorizes RSOs into mission types	More objects observed, increased ability to track, quicker ability to identify
Characterization	Captures the features and employment of an RSO by integrating information from sensors to support space operations and assess threats	Improved data integration and exploitation, quicker analysis
Threat warning and assessment	Assesses potential threats to space operations and evaluates their impact on an RSO's mission and ability to deliver effects	Enhanced anomaly detection, improved analytic capabilities, improved data integration and exploitation
Data integration and exploitation	Exploits multiple data sources and fuses them to support decisionmaking, providing higher confidence and earlier predictions	Quicker integration and exploitation, improved data quality and confidence

SOURCES: Discussions with SDA stakeholders; Joint Publication 3-14, *Space Operations*; USSF, *Operations*.
NOTE: RSO = resident space object.

SDA Operators

Two squadrons function as SDA operators. The 18th SDS is a component of the USSF Space Delta 2, located at Vandenberg Space Force Base. The 18th SDS is tasked with executing the USSPACECOM SDA mission and processing SSN data to monitor activity in space and maintain custody of RSOs. The 18th SDS does so through the aggregation of observations from the SSN: calculating current and predicted orbital states for RSOs, conducting launch detection, tracking, and

[14] Joint Publication 3-14, *Space Operations*; USSF, *Operations*.

identifying potential collisions. A key piece of this mission is the maintenance of a comprehensive catalog of satellites.[15]

Activated in 2022, the 19th SDS is also a component of Space Delta 2 and has assumed the mission of CA from the 18th SDS. Further, the 19th SDS serves as a primary backup to the mission functions at the 18th SDS.[16] Together with the 18th SDS, the 19th SDS maintains and populates the publicly available space catalog Space-Track.org.[17] The squadrons' primary mission functions are noted in the box below.

Primary Mission Functions for 18th and 19th SDSs

USSF Space Delta 2: Prepares and presents combat-ready SDA operations to assigned and attached forces

18th SDS: Squadron within Space Delta 2 tasked with tracking and monitoring space objects, collision prediction, maneuver detection

19th SDS: Squadron within Space Delta 2 tasked with backup to 18th SDS functions, CA, beyond geostationary earth orbit (xGEO) SSA

Combined Force Space Component Command: USSPACECOM subordinate command responsible for the management of sensors used to conduct SSA

Joint Task Force–Space Defense: Protects and defends U.S. space assets, working with the intelligence community through the National Space Defense Center SDA units

SOURCES: USSF, "18th Space Defense Squadron"; Peterson Schriever Space Force Base, U.S. Space Force, "19th Space Defense Squadron"; U.S. Government Accountability Office, *Space Situational Awareness*.

The Changing SDA Mission and Infrastructure

Meeting the challenges of the SDA mission requires higher speed to respond to operational needs in a crisis or conflict, scalability as the number of space operators grows, and more-complex analyses to address increasingly maneuverable RSOs and the growing numbers of both cooperative and noncooperative space vehicles. Evolving the SDA mission introduces a variety of challenges.[18]

[15] Wilson and McKissock, "18 SDS Small Satellite Support." There may be a shift in the burden for CA services on USSF Space Delta 2 in the coming years, but the growing SDA mission will likely outpace the relief on analyst demands provided by this shift. In 2018, Space Policy Directive–3, "National Space Traffic Management Policy," directed the U.S. Department of Commerce to take over SSA and space traffic management services. However, the path and timeline for doing so remain unclear.

[16] USSF, "18th Space Defense Squadron."

[17] 18th and 19th SDSs, Combined Force Space Component Command, *Spaceflight Safety Handbook for Satellite Operators*.

[18] Although the SDA functions described above are still relevant, the overall SDA mission is evolving as space becomes increasingly congested, contested, and competitive (DoD and Office of the Director of National Intelligence, *National Security Space Strategy*, p. 1). These changes are introducing a variety of challenges to operators responsible for providing SDA—for example, keeping up with the pace of data ingestion and processing for SDA, with the ability to provide analysis in support of increasingly complex RSO behavior and with the ability to scale in response to increasing numbers and maneuverability of space objects. Gen Saltzman has acknowledged the importance of addressing these timing and analysis challenges: "When I hear about

The major limiting factor is the USSF's reliance on legacy systems to perform the SDA mission.[19] One challenge is the need to augment legacy systems with modern tools, methods, and processes. These legacy systems were designed for a mission that focused on cataloging objects rather than supporting a warfighting domain.[20] Another challenge is turning the SSN sensor data into actionable information, which requires computers and software to integrate, manage, and analyze these data. The SSN is mostly composed of aging sensors, which were designed during the Cold War to address threats from nuclear weapons. Finally, upgrading computing and software must be done without disrupting the ability to continue to monitor, detect, and assess nuclear threats or the SDA mission.

There are several important efforts within the SDA mission to better address these challenges. The efforts are particularly important as we examine where AI/ML tools can have impact because the limitations and challenges of the current process and architecture might not be the limitations and challenges of tomorrow's process and architecture. Therefore, what this future architecture looks like may shape how AI/ML tools can affect this mission, and if we are deciding where to implement AI/ML tools, we need to decide whether we are designing for the present or for the future.

Changes in SDA Infrastructure

Legacy systems must be replaced in light of growing SDA needs. The existing systems—SPADOC and Command, Analysis, Verification, and Ephemeris Network (CAVENet)—have been challenged for years now with keeping up with the growing satellite catalog.[21] To replace the overwhelmed SPADOC system, the USSF has funded L3Harris to develop the Advanced Tracking

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a breakup that occurred of a rocket body, where one rocket body became five pieces of rocket body, but it took us a couple of days to put all that information together—okay, that's probably not the kind of timeline that would allow the Space Force to take action" (quoted in Luckenbaugh, "Space Symposium News").

[19] The recognition of this need is not new, and finding a way to modernize or replace these systems has plagued DoD since the 1980s, with these attempts facing challenges to stay on schedule and budget. See, for example, Albon, "US Space Force Preparing to Decommission Legacy Command and Control System." Attempts to modernize Space Defense Operations Center (SPADOC) began in the 1980s. Notably in 2009, the Joint Space Operations Center Mission System was yet another attempt to replace SPADOC, but only one of three proposed increments was fielded, and the program was well behind schedule and over budget. Further discussions of these challenges are reported in U.S. Government Accountability Office, *Space Acquisitions*.

[20] Warfighters need to pivot from the historical SSA approach that focuses on space catalog maintenance to one that can be operationally responsive in a congested and contested environment, as well as one that can assess intent and characterize behavior. This means that SDA needs to provide decisionmaking support on a relevant timeline, as well as anticipate problems and intent.

[21] SPADOC was brought online in 1979 in the Cheyenne Mountain Complex as a cataloging system, and its last significant upgrade was in 1989 (Clark, "What About JMS?"). Gen Jay Raymond, then–head of Air Force Space Command, said in a speech in 2017: "I can't wait until we can take a hammer to SPADOC and just blow it to bits. It's an old clunker and it's a catalog system: it's not a warfighting command and control system" (quoted in Clark, "What About JMS?"). This prompted the USSF to develop CAVENet in the 2000s to augment SPADOC's capabilities and provide data analysis and sharing capabilities on a local network (Hitchens, "Key Space Monitoring Sensors Still Rely on Outdated CAVENet Computer System"; see also Weeden and Cefola, "Computer Systems and Algorithms for Space Situational Awareness," for a more in-depth accounting of the development of computer systems and algorithms for SSA). CAVENet is an offline system of workstations that run the Astrodynamics Support Workstation (ASW) software, which provides analytic tools for classified historical analysis, such as collision and threat assessment (Jordan Logue, discussion with representatives from the 18th SDS, April–July 2023; Hitchens, "Key Space Monitoring Sensors Still Rely on Outdated CAVENet Computer System"). However, as SPADOC became increasingly unable to keep up with the growing SDA mission, CAVENet has taken on operational tasks it was never designed for: CAVENet now keeps a separate satellite catalog of special perturbation (SP) vectors and covariance for analysis, and CAVENet is now responsible for developing the daily SSN tasking plan.

and Launch Analysis System (ATLAS), a software platform for monitoring launches, satellites, and debris. ATLAS is intended to allow for the decommissioning of SPADOC and provide SDA and command and control (C2) capability to the 18th SDS and the National Space Defense Center.[22] ATLAS development is managed by the USSF Space C2 program, nicknamed Kobayashi Maru.[23]

Warp Core and the Unified Data Library (UDL) are two recent efforts to modernize data management for the USSF. Warp Core, co-managed by SSC and Kobayashi Maru, is developed by Palantir to provide a common data interface, facilitate data sharing, and support decommissioning legacy systems.[24] The UDL, managed by SSC's Cross Mission Data team, is a cloud-based data repository that provides data management support and facilitates the integration of multiple datasets, which Warp Core can then curate and fuse.[25]

Significance of SDA Process Changes

The implementation of AI/ML tools to enhance SDA mission effectiveness depends on whether new systems, such as ATLAS, come online to replace SPADOC, because these new systems could expand processing and analytic capabilities and provide an interface for AI/ML tools. **Continued reliance on legacy data processing systems (SPADOC and CAVENet) limits the benefit of improving on-orbit sensing technologies.** For example, Space Fence and the Geosynchronous Space Situational Awareness Program (GSSAP) are two systems that can track satellites or debris. However, both systems must route data through CAVENet before entering Warp Core.[26] More-direct feeds to Warp Core would eliminate the need for CAVENet and open the door, potentially, which could improve SDA mission effectiveness through enhanced sensing technologies.

An important element of these process changes will be the ability to test and validate new AI/ML tools. Software sandboxes are necessary for building and implementing new tools and capabilities, including AI/ML capabilities, for the SDA process. Integrating AI/ML tools into systems that support SDA will first require a software platform on which to develop, test, and integrate new applications.[27] ATLAS intends to provide this platform, but there are significant challenges that need to be addressed to realize this needed replacement of the aging software and

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[22] See, for example, Erwin, "Space Force Extends L3Harris' Contract to Upgrade Space Tracking System."

[23] Kobayashi Maru was named after a training exercise in the television show *Star Trek* designed to represent a no-win scenario. Only James T. Kirk was able beat the scenario by changing the code, and this is the inspiration for using this name to represent the objective of doing things differently and changing the code for the space C2 mission. See Krolikowski et al., "Space Command and Control Program—Kobayashi Maru."

[24] SSC, "Warp Core."

[25] Han and Sodders, "SSC Data-Management Software Plays Critical Role in SDA, Afghanistan Airlift"; SSC, "Warp Core."

[26] Space Fence links to CAVENet via the Non-Traditional Data Pre-Processer (NDPP). GSSAP links to CAVENet via Red LAN, a distribution node for SDA sensors that can deliver at multiple classification levels. Warp Core pulls the data from CAVENet and sends them to ATLAS (Hitchens, "Key Space Monitoring Sensors Still Rely on Outdated CAVENet Computer System").

[27] Another such software development program is Defense Readiness Agile Gaming Online Network (DRAGON), an Air Force Research Laboratory program with a 14-day sprint cycle for operational and developmental test and evaluation. This program provides software development in support of Joint Task Force—Space Defense Commercial Operations and Joint Task Force—Space Defense operations. See Bonnette, "JCO Executes DRAGON Army Ops Days to Ensure Commercial Space Integration."

hardware.[28] However, our interview with the Kobayashi Maru team revealed a willingness and a budding capability to test new capabilities within sandboxes.

Investing in capabilities that will be most effective in the current architecture might not produce the highest impact in the future architecture. Because the SDA mission is changing and growing in complexity, SDA capabilities need to be able to scale and morph to meet these changes. Areas for impact therefore need to at the very least be able to scale—and ideally would facilitate this scaling while also enabling more-complex analysis and processing. The above efforts are setting the stage for more AI/ML tool integration while allowing for the decommissioning of legacy systems.

Because of these anticipated changes, the development of AI/ML tools for the SDA mission needs to be guided by the infrastructure of tomorrow. Meeting today's challenges with AI/ML tools and meeting the likely challenges of tomorrow require different technology. Without the decommissioning and replacement of legacy systems, SDA mission effectiveness would likely be limited by processing and analytic capabilities and capacity. With the replacement of SPADOC (and CAVENet) with ATLAS or a similar program, there may be more processing capabilities available, allowing a shift in focus to refining the types of analysis that can be performed.

Infrastructure changes may enable higher impact from AI/ML tool implementation. The proposed change in infrastructure supports the validation of additional tools, potentially increasing the pace at which new AI/ML tools could be implemented. Reducing the data transport time between the 18th and 19th SDSs would eliminate a bottleneck that would in turn allow for any other process speed improvements to be realized.

Chapter Findings

It is important to note that the SDA mission is undergoing significant changes due to the increase in launches and maneuverable RSOs. The USSF is faced with the challenge of adapting to the changing operational environment with limited resources that were not originally designed to address these threats, although modernization efforts are currently underway. In the next chapter, we provide a detailed account of how existing SDA processes are being used to address these challenges by documenting the flow of information from sensors to decisionmaking. We identify potential bottlenecks and explore the potential for AI/ML tools to improve the efficiency and effectiveness of these processes.

[28] ATLAS is the logical platform to support and host AI/ML tools. However, its first priority is to come online and decommission the legacy systems it is designed to replace. Therefore, AI/ML is potentially on the ATLAS road map but not a current priority. See Hitchens, "Key Space Monitoring Sensors Still Rely on Outdated CAVENet Computer System."

Chapter 3

Applying Artificial Intelligence in Space Domain Awareness

AI is a broad term that lacks a complete and commonly accepted definition. We also note that AI and ML are often perceived as interchangeable but, though related to another, are distinct. A comprehensive discussion of AI taxonomy is not within scope of this report, and, for the purposes of this report, we use the term *AI/ML* inclusively when referring to concepts within this taxonomy. We avoid definitional debate and consider AI/ML as a set of technologies that broadly provides six capabilities to the USSF: computer vision, language processing, reinforcement learning, prediction and classification, generative learning, and expert systems.[29] These capabilities are typically used to improve four broad areas: efficiency, efficacy, resiliency, and human capital metrics.[30] We consider how incorporating AI/ML into SDA can yield improvements in these areas in service of the SDA mission. AI/ML applications can vary, but there is typically a complexity and automation trade-off. The most-straightforward applications of AI/ML are to automate well-defined and less complex tasks that are tedious for a human or to provide suggestions to support decisionmaking.

We turned to developing a detailed look at the space catalog and CA processes for their foundational role in other SDA processes. The SDA mission is ultimately tasked with building an *understanding* of the space environment, activities, and behavior of RSOs to support decisionmaking. This understanding comes with producing *knowledge*, the fundamental output of SDA, and is accomplished by characterizing what *has* happened as reported via sensor observations to anticipate what *might* happen and to support better course-of-action development and timely, and *wise*, decisionmaking. Figure 3.1 provides a view of these relationships and highlights that these bottom tiers of this hierarchy enable higher tiers. To build this knowledge requires the ability to pull in data and then synthesize, integrate, and analyze these data to produce information. Improving SDA requires better D/T/ID and characterization; that is, to improve knowledge of what is going on in the space domain, we need to focus on creating information out of data and using information from multiple sources.[31] Therefore, this ability to use SSN and other data sources to characterize space

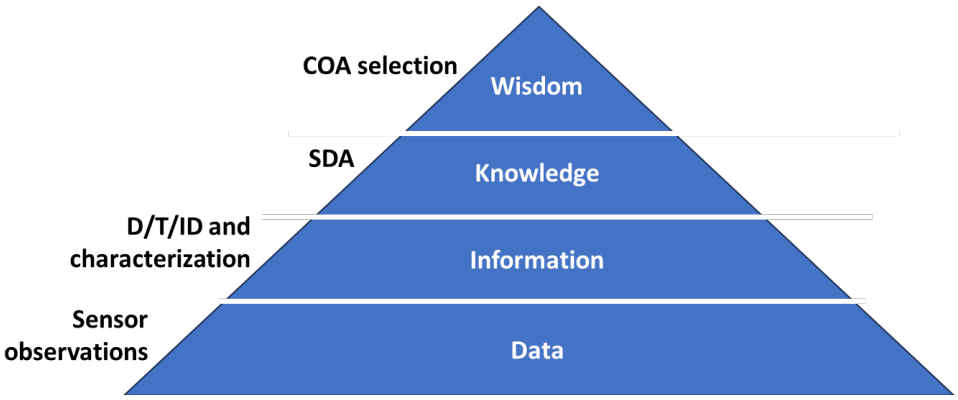
[29] Different academic fields have competing definitions of *AI* and *ML*. Refer to Chapter 1, which provides *AI* and *ML* definitions from U.S. Code, Title 15, Section 9401. We focus on understanding what automation, AI, and ML capabilities can help the USSF in the SDA mission. Therefore, we consider AI/ML as one broad category that provides capabilities. These capabilities follow the standard text of Russell and Norvig, *Artificial Intelligence*, with additional insight from Wehbe and Ramdas, "Introduction to Machine Learning."

[30] Menthe et al., *Technology Innovation and the Future of Air Force Intelligence Analysis*: Volume 1, *Findings and Recommendations*, and Menthe et al., *Technology Innovation and the Future of Air Force Intelligence Analysis*: Volume 2, *Technical Analysis and Supporting Material*, include a series of "measures of effectiveness" to evaluate how novel technologies can improve U.S. Air Force intelligence processing, exploitation, and dissemination.

[31] Rowley, "The Wisdom Hierarchy."

activities is foundational to SDA, and we focus in particular on CA because of its embodiment of these elements.

Figure 3.1. Hierarchy Illustrating Functional Relationships for SDA and Space C2



SOURCE: Adapted from Ackoff, "From Data to Wisdom."
NOTE: COA = course of action.

SDA is a uniquely challenging and technical mission set that grows in complexity while personnel, computer, and sensor resources stay relatively static. This challenge is felt acutely in the CA process as more RSOs need to be identified, tracked, and assessed. SDA occurs in eight-hour cycles: Approximately the first 40 minutes are spent on updating the High Accuracy Catalog (HAC) (via the special perturbations model) and the rest of the time are spent on the CA process.[32] Any improvement within the CA process frees up computational and analytical bandwidth for other SDA tasks.

We assess each aspect of the CA process to identify several key areas for process impact. Proposed tools to address these needs range from tools that facilitate an increase in frequency or speed for existing processes, incorporate new processes, or replace human tasking with AI/ML tools to free up analyst time. AI/ML tools are useful as force multipliers to address resource limitations faced by operators due to aging systems, increasing mission demands (and operations tempo), and personnel shortages. These tools can help operators maintain SDA even as the mission grows in size and complexity.

Existing Efforts

There are significant efforts already underway to develop AI/ML tools to support the SDA mission. We highlight some below, but this is not meant as an exhaustive list:

- **Increasing number of sensor observations:** Work to support increased detection and improved accuracy of sensor observations is ongoing, including efforts within the Space

[32] Within 40 minutes, the HAC is updated. The next two hours are spent transferring the data updates from the 18th to the 19th SDSs. Using the HAC, Automatic Conjunction Assessment (AutoCA), and Super Computation of Miss Between Orbits (SuperCOMBO) (described in later sections) take up all of the computational capacity on CAVENet. Jordan Logue, discussion with representatives from the 18th SDS, April 2023.

Systems Command SDA Acquisition Delta (SSC/SZG) AI and Autonomy Sandbox colocated with the Space Delta 2 15th Space Surveillance Squadron. Ongoing work here has focused on developing computer vision algorithms to improve the detection rate and precision for satellite observations in the Machine Learning for Space Superiority (MISS) program.[33] The SensorCapsule program at SSC/SZG, for example, is enabling the increased use of commercial sensing data by helping connect nontraditional sensors to the UDL.[34] Improved sensor use would also support increasing the number of observations, and the SSC/SZG mission-driven, autonomous, collaborative, heterogeneous, intelligent, network architecture (MACHINA) represents ongoing work on sensor orchestration in support of sensor management optimization.[35] To deconflict from this ongoing work, we did not focus on planning capabilities in our research.

- **Ensuring that tools can be tested and validated:** SSC/SZG is developing sandboxes for AI/ML tools as part of the Pivot SDA portfolio. This sandbox is configured to interface with UDL data and enables individual providers to develop analytic tools that suit their specific needs on a government-owned platform, removing barriers to third-party development of tools and allowing more-rapid deployment of models. This capability was demoed with commercial sensing data in a recent Sprint Advanced Concepts Training (SACT) exercise in 2022.[36]

Sensor Tasking, Scheduling, and Prioritization

The 18th SDS coordinates the observation activities of the different sensor sites composing the SSN. There more RSOs spread across too much sky than the 18th SDS can observe every day, so the 18th SDS must stagger observations to capture every object with sufficient frequency to correlate with each's track history. Additionally, different sensor sites feature different technological configurations, reliabilities, percentage downtimes for maintenance, and precisions; these factors all must be accounted for as part of the sensor tasking process. *Sensor allocation* refers to the deliberate distribution of taskings to individual sensor sites to achieve the twin goals of catalog maintenance and event capture. To maximize the number of individual observations and minimize uncertainty in orbit determination (OD), it is necessary to optimize the allocation of these limited sensor resources. The 18th SDS orbital analysts are responsible for building the daily composite (COM) file, which is a composite of tasking orders for all sensor sites, also known as the *consolidated tasking list*.

Issues

The 18th SDS has a Sensor Optimization Cell that monitors SSN sensor site performance. Performance metrics, which are based on laser calibrations, are constantly produced at each site.

[33] Fletcher, "AI and Autonomy for SDA."

[34] Fletcher, "AI and Autonomy for SDA."

[35] Fletcher, "AI and Autonomy for SDA."

[36] Fletcher, "AI and Autonomy for SDA."

However, the 18th SDS is resourced to check these metrics only periodically. Sensors that are out of calibration can produce erroneous data about a particular target, limiting the squadron's ability to characterize high-interest behavior and polluting the space catalog.

Tracking RSOs is ultimately an uncertainty (referred to as *covariance*) reduction problem: Covariance grows over time (from maneuvers or changes to atmospheric density solar radiation) and can be reduced only by sufficiently accurate sensor observations. Sensor allocation is a balance between uncertainty reduction and revisit rate using limited sensor resources. Unfortunately, the current paradigm emphasizes revisit rates. The daily taskings are often overridden in favor of higher-level guidance to track priority RSOs. As a notional example, guidance may require that a specific RSO needs to be observed *X* times within a day. To meet this requirement, the 18th SDS may task a sensor that is unable to get the needed collection (e.g., an optical telescope currently facing the sun or a radar with higher covariance than the current object's covariance). The opportunity cost of a tasking override is missed detections, which may lead to uncorrelated RSOs (uncorrelated tracks will be discussed in the "Catalog Maintenance and Uncorrelated Tracks" section later in this chapter).

SP Tasker was not designed to handle the large quantity of objects currently in orbit and is at the limits of its abilities. Nor was it ever designed to support agile responses to changing real-time observation needs. The increases in the population of RSOs and frequency of high-interest events pose a challenge to SP Tasker's processing capabilities and operations cadence.

The upcoming ATLAS program, as described in Chapter 2, is intended to replace SPADOC. In addition to having more processing capacity, ATLAS could potentially augment the SSN using information from non-SSN U.S. government sensors; it remains an open question as to how to optimize load balancing across these capabilities and how to weigh and trust information from non-SSN sensors.

Potential AI/ML Application

Sensor calibration is a major challenge for the 18th SDS. Because of our stakeholder interviews, we believe that the 18th SDS wields the responsibility for tasking sensor sites to perform laser calibrations, processing the resulting calibration data, and directing the adjustments for improving subsequent observations from that site. Given the acute manpower crunch at the 18th SDS, solutions include either (1) empower sensor sites to perform their own calibrations as needed or (2) automate calibration tasking. The first option requires additional training and systems for sensor site staff but reduces the workload of the 18th SDS. The second option requires the development of automated calibration tools (e.g., calibrate at regular intervals or set thresholds to identify anomalous observations).[37]

Either option will provide the 18th SDS with an improved ability to track sensor covariance over time, which can lead to improved sensor use. A separate tool can track these attributes for each sensor site and determine feasibility of potential taskings and optimize sensor allocations based on timeline requirements (to capture a potential high-interest event) or prevent an RSO from becoming lost

[37] To the best of our knowledge, calibrating observations is a manual process today.

(thereby creating an uncorrelated track and analyst object). These features are enabling steps toward dynamic sensor retasking.

Automating sensor calibration is a first step toward creating a sensor validation process. As the USSF considers leveraging non-SSN capabilities (U.S. government or commercial) to augment the aging SSN sensors, creating a framework or support to validate sensors smooths the transition toward ingesting new sensor data and new data formats.

Data Collection and Integration

Once SSN sensor observations are taken and transmitted to the 18th and 19th SDSs, the second stage of data integration begins. Sensors observe range, range rate, azimuth angle, and elevation angle and send the data to the SPADOC Space Data Server housed at Cheyenne Mountain via data cables, where they are translated into a format that is ingested by CAVENet. Observation data across each sensor site must be fused because sites are not identical. The current fusion procedures are complex and put CAVENet at its maximum processing capacity. Note that the 18th SDS must access these data through CAVENet, but the 19th SDS has direct, albeit slow, data lines to the SSN because of the legacy Navy Space Surveillance System.

Issues

The number of RSOs is steadily growing, but processing capabilities and the number of SSN sensors are relatively static. To meet this expanding need, it is vital to leverage additional sensing resources, including commercial sensors. Success will entail effectively ingesting, fusing, and validating data sourced from disparate resources. Processing power and data communication are the two primary challenges in this stage, and the new ATLAS system is intended to address these bottlenecks with both hardware and software upgrades.[38] The Innovation Cell within the 19th SDS is also actively working on these issues.[39]

The current SSN's capabilities do not match the new reality of near-daily launches and mega-constellations of hundreds of satellites. This problem is exacerbated with the increase of noncooperative space entities, those domestic and international O/Os who do not voluntarily provide the 18th SDS with screening ephemerides. Mega-constellations, in particular, have caused a drastic shift in operational tempo.[40]

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[38] The SSN is old, and the centralized system, SPADOC, has coordinated the SSN since the 1970s and relies on hard data lines for all communications with the sensor sites. Its replacement, ATLAS, is under development, but its development and uptake have been negatively affected by the failure of the last effort to replace SPADOC: Joint Space Operations Center Mission System.

[39] Li Ang Zhang, George Nacouzi, and Melissa Heron, in-person visit to the 19th SDS, December 2022.

[40] In response to mega-constellations, the 18th SDS conducts these activities:

- standing up the new Mega-Constellation Analysis Cell dedicated to cataloging constellations with more than 100 satellites
- disambiguating objects during large-scale deployments (this activity is difficult because of proximity and is possible only with O/O-provided or commercially procured ephemerides)

Potential AI/ML Applications

New sensor modalities—such as on-orbit observations, passive radio frequency, or commercial sensors—can enhance SSN observational data and help address accelerating space activity. The 18th SDS needs support for ingesting new sources of sensor data, in addition to automating sensor validation. Although computational limitations may limit this capability today as a result of the reliance on legacy infrastructure for data transmission and analysis, the inclusion of data from additional sensors that are not part of the SSN is something that DoD is actively pursuing—for example, through the Joint Task Force—Space Defense Commercial Operations cell and could be supported through the use of AI/ML tools.[41]

Academic research indicates that computer vision may help disambiguate individual RSOs within a mega-constellation and mimic C2 behavior based on existing constellation management.[42] Data fusion is an evolving practice, and AI/ML can enable new methods of observation and potentially even new prediction models.

OD

Following data collection and integration, the 18th SDS uses OD to determine the most likely state of a given RSO from sensor observations. Each sensor site takes multiple observations of a target RSO, and there are multiple sources of uncertainty in these observations, including sensor noise and bias and RSO motion uncertainty. To mitigate the impact of uncertainty, the 18th SDS uses the weighted least squares method to determine the most likely state for the RSO.[43] Each observation, weighted by its confidence (inverse of sensor covariance),[44] influences the most likely RSO state.

These observations, and the associated uncertainties, feed a process known as *differential corrections* that estimates the most likely *averaged* state of the object at a given time: a so-called *ephemeris*. Time series of ephemerides are recorded in the HAC maintained by the 18th SDS as one of the squadron's core missions. The HAC serves as the starting point for forecasting future RSO trajectories and performing CA. Improvements in OD, such as faster processing, more-frequent observations or more-accurate observations, will all improve downstream processes.

- employing ion thrusters in mega-constellations to maintain their orbits; the frequent station-keeping burns result in non-Keplerian orbits that violate the elliptical orbit assumptions used for OD calculations
- tracking the growing orbital population with the SSN, which was designed for ballistic missile warning; senor sites are not necessarily located facing modern satellite population centers.

[41] The Joint Task Force—Space Defense Commercial Operations allows for the use of commercial SSA data to support DoD. See, for example, U.S. Government Accountability Office, *Space Situational Awareness*.

[42] Massimi, Ferrara, and Benedetto, "Deep Learning Methods for Space Situational Awareness in Mega-Constellations Satellite-Based Internet of Things Networks"; Kopacz, Roney, and Herschitz, "Deep Replacement."

[43] Because of computational constraints and historical reasons, CAVENet uses weighted least squares estimation to update the catalog with new observations.

[44] The uncertainty associated with a sensor observation is quantified as *covariance*, the matrix of values describing the spread of each state variable (e.g., x, y, and z directions in a Cartesian system) with respect to itself and every other state variable. The spread of values for a given observation depends on such sources of uncertainty as sensor noise and bias.

CAVENet ASW's Automatic Special Perturbations Ephemeris Generation (ASP) script is run regularly to produce SP propagations for the updated ephemerides,[45] after which the determined trajectories are transferred over to the 19th SDS via dedicated data lines through Cheyenne Mountain.

Issues

OD's later stages involve propagating an object's most recent ephemeris and covariance into a different time (either into the future for CA or into the past for object association). This step is resource intensive and is currently performed only by the 19th SDS on a subset of RSOs because of practical processing limitations.[46] The RSO covariance propagation has the potential to provide two benefits.

First, propagating an RSO's future covariance would enable the 18th SDS to proactively identify objects that are about to be lost during the Automatic Differential Correction (ADC) process.[47] Currently, this is a reactive process in which a percentage of daily observations fail ADC and must be recorrelated manually. These objects can be very difficult to recorrelate and may take anywhere from minutes to days, if at all.[48]

Second, RSO covariance is a critical aspect of improving sensor tasking. When tracking an RSO, there are two sources of covariance: the object and the sensor. To make a useful observation, the selected sensor must have lower covariance than the object; otherwise, the observation may actually contribute to object covariance. Tracking both RSO and sensor covariance can improve the sensor allocation process and enable a feasibility calculation (what *can we improve* with detections today?). A feasibility assessment might also mitigate conflicts between SP Tasker and revisit rate mandates for high-interest RSOs.[49]

Potential AI/ML Application

Most downstream components of SDA depend on accurate OD. This is a step that can be appropriate for an AI/ML application. Neural networks are universal function approximators, and academia is demonstrating that they can feasibly approximate complex physics models.[50] Although neural networks will typically yield less accurate solutions (relative to a known physics model), they

[45] ASP currently takes about 40 minutes and runs three times per day at the beginning of each eight-hour cycle.

[46] The 19th SDS propagates refined RSO ephemerides and covariances for objects that were flagged for risk of collision (as calculated by a CA prescreening algorithm). These covariances are used to calculate a more authoritative collision probability.

[47] Li Ang, Krista Langeland, George Nacouzi, and Jon Tran, discussion with various analysts at the 18th SDS, April 2023; Li Ang Zhang, George Nacouzi, and Melissa Heron, discussion with various analysts at the 19th SDS, December 2022.

[48] Li Ang, Krista Langeland, George Nacouzi, and Jon Tran, discussion with orbital analysts at the 18th SDS, April 2023.

[49] As discussed earlier, there revisit-rate mandates often override SP Tasker. The result may be improper sensor tasking for the sake of meeting the mandate. Tracking both RSO and sensor covariance enables a feasibility calculation: Although we *can* achieve a revisit rate, will it *improve* our situational awareness (in terms of decreasing uncertainty on an RSO)?

[50] Hornik, Stinchcombe, and White, "Multilayer Feedforward Networks Are Universal Approximators."

can achieve high-accuracy estimates at a fraction of the time. This speedup can be a highly desirable aspect for SDA.

We explored whether any element of OD can be approximated by a neural network and whether a neural network offers any benefits compared with status quo processes. To accomplish this, we developed in-house models to rapidly prototype AI experiments and determine application viability. The primary AI capability we leveraged was prediction (regression) to train a neural network to closely align inputs and outputs with a physics-based numerical orbit model. We developed two neural network models: one to propagate an RSO state into the future and one to propagate time-varying covariance into the future.

Using standard orbit propagation parameters and an RSO's initial state, we leveraged Poliastro,[51] an astrodynamics Python package, to generate ground truth trajectories and a standard architecture neural network to approximate inputs and outputs. On the basis of our testing, we determined that **state-only propagation with AI/ML is feasible**. Within 90 milliseconds, the neural network is able to project an RSO state ten days out with a root mean squared error (RMSE) of 1 km.[52] For context, the neural network is able to propagate a 40,000-sized space catalog in one hour. By our best estimates, OD takes several hours for a similar sized-catalog (due to processing constraints at the 18th SDS) and the general perturbation catalog is accurate to several meters (<1 km). The accuracy of the neural network can likely improve, but this invokes a speed-versus-accuracy trade-off. Additional details and discussions are provided in the companion report, which presents the results from two AI/ML case studies.[53]

State-only OD offers several advantages for SDA:

- **Analysis is faster**, and orbital analysts within the 18th SDS spend significant time associating uncorrelated objects to known objects (e.g., breakup events or lost objects). Fast, near-instantaneous ten-day state propagation can help analysts quickly assess potential matches or filter out unlikely candidates. Quicker and more-frequent calculations could in turn promote a more responsive (i.e., more-frequent) OD process, which would support the maintenance of a more current HAC for screening efforts at the 19th SDS.
- **Conjunction prescreening can be sped up**, minimizing the number of false-positive RSO pair candidates that require high-fidelity calculations. State-only propagation may speed up elliptical volume screening (see the "CA Calculation" section later in this chapter for more information).

We trained a second neural network to propagate an RSO's covariance. For this experiment, we used Bayesian neural networks (BNNs), an architecture designed to replicate probability distributions, to propagate an RSO's covariance into the future. However, our results showed that this approach is significantly more difficult than the state-only propagation network. It is difficult to contextualize the accuracy of the BNN because we lack an understanding of the operational requirements of current covariance propagation models, but the BNN did not fully replicate time-

[51] Poliastro—Astrodynamics in Python, homepage.

[52] The 90-millisecond figure is based on our Nvidia Tesla P6 graphics processing unit (GPU) workstation. This likely can be accelerated even further with a high-performance computer or with more-modern GPUs.

[53] Tran et al., *Artificial Intelligence and Machine Learning for Space Domain Awareness*.

varying covariance in our test data.[54] Although we could not demonstrate BNN feasibility to propagate covariance,[55] with operational requirements and sufficient training data, such a tool may one day offer a faster alternative to high-fidelity CA screens.

Catalog Maintenance and Uncorrelated Tracks

Catalog maintenance involves updating the HAC via OD and the association of uncorrelated tracks (UCTs).[56] UCTs are those strings of observations that are unable to be automatically associated with an object in the catalog. This may be because the observed object has been lost and has been assigned to an analyst object with no track history, or it could be because the object's orbit has changed so significantly that the system is unable to recognize its relationship with a past track. There are numerous potential causes, including maneuvering or fragmentation events. During OD, the ADC process successfully associates the majority of observations to a known object, but failure can occur when new observations cannot be associated with an existing track within set tolerance thresholds.[57] These failures trigger a manual review for analysts within the 18th SDS.

Issues

The orbital environment is rapidly evolving, with increased launch frequency, increased RSO maneuvering, and the proliferation of noncooperatives throughout the orbital environment. These events lead to ADC failures at a pace that is outstripping the 18th SDS's current analyst ability.

UCTs are assigned a SPADOC code for future bookkeeping. These so-called analyst objects currently require manual intervention by 18th SDS analysts for reassociation with their past known custody. The longer analyst objects go without reassociation, the more time this process takes and the less likely it is to be successful. A UCT may become lost and reappear as a separate UCT; a competitor may be able to take advantage of this limitation to obscure deployment of spacecraft.

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[54] *Accuracy* is not precisely used here. To evaluate the outputs of the BNN, we used different evaluation metrics, such as an output's mean average percentage error (MAPE) from the mean of the ground truth covariance and Kullback-Leibler divergence of the output's distribution relative to ground truth distribution. However, this is difficult to contextualize because we were unable to assess the operational performance of ASP during this study and therefore could not compare these metrics against operational reality.

[55] We attempted to troubleshoot the issue by adjusting various parameters of the BNN, but none of these adjustments resulted in improved performance. We also considered the possibility that our dataset was insufficient or that there were issues with the quality of the data.

[56] We also refer to these objects as *lost objects*.

[57] A set of observations is assigned to the closest existing track. ASW verifies this assignment for a solution precision threshold, quantified by weighted root mean squared (WRMS) error. If the WRMS error of a Gaussian batched-weighted least squares solution is too high, the ADC system will reject the solution and flag the observations for manual inspection.

Potential for AI/ML Tools to Support Catalog Maintenance

Object correlation appears to be a key area for AI/ML impact. Although the vast majority of observations are automatically assigned, the increasing numbers of noncooperative space entities and total RSOs increase the amount of UCTs.

Some aspects of correlating UCTs can be automated, **enabling more-frequent object association**. Orbital analysts within the 18th SDS follow a highly complex step-by-step task list to reassociate UCTs. List generation processes are typically undocumented and depend on the experience of the individual being tasked. We recommend codifying these tasks to preserve the knowledge and potentially automating these rules into an expert system of *if-then-else* rules. This tool would be able to handle the most-common scenarios for ADC failure without requiring human intervention. Not every UCT can be correlated this way, but such a system can help shrink the number of UCTs and **free up analyst bandwidth** for more-difficult scenarios.

This system can be augmented by compiling a database of previously unknown analyst objects and their eventual reassociation to known tracks. Using the expert system as a starting point, an advanced AI/ML application can be trained to recognize the relationships in UCTs and generate automatic track association recommendations. Splitting the development of this AI/ML application into layers of increasing complexity can mitigate development risks.

CA Calculation

CA is the probabilistic process by which the 18th and 19th SDSs quantify risk of collision in space. The process is probabilistic rather than deterministic because of the numerous and chaotic orbital forces that progressively perturb an object's trajectory away from its theoretical path over time. Because the closing velocities involved in space collisions necessitate avoidance maneuvering far in advance of the predicted time of closest approach (TCA), it is (currently) impossible to know *exactly* where the two colliding objects will be at the TCA. CA thus requires a dynamic assessment of risk that supports continuous ingestion of new information to improve confidence in the prediction and avoid unnecessary maneuvering.

Probability of collision (P_c) is a measure of confidence of a predicted collision occurring. Calculations of P_c seek to minimize the number of false positives by accounting for the dynamic uncertainties built into the RSOs' respective state space variables.[58] Increasing P_c accuracy is thus a key objective for CA; to achieve it, the 18th and 19th SDSs leverage their OD and sensor tasking processes to aggregate raw sensor observations for bookkeeping in the space catalog and subsequent calculations of P_c. To characterize these processes, we consulted with analysts to identify the decision elements embedded within that support the timely and accurate determination of P_c.

The CA process flags pairs of RSOs that are at high risk of collision. In the early days of space operations, operators held that the space domain was so voluminous that the odds of collision were remote—the Big Sky assumption. Subsequent collisions demonstrated the invalidity of this

[58] Note that the need for uncertainty quantification in the form of a covariance matrix, propagated along with the state vector itself, necessitated the development of SP methods (whose numerical nature would allow incremental bookkeeping of uncertainty in a way that general perturbation could not. The current HAC of RSOs has been maintained according to SP ever since.

assumption, as well as the outsized impact of even a single collision on the celestial environment, leading to a recognition of the need to assess the risk of collision in space. The 19th SDS is primarily responsible for CA. Using the latest RSO states from the HAC, the 19th SDS calculates CA using two ASW tools: AutoCA and SuperCOMBO. ASW's ASP uses the SP model for OD and propagates all HAC object states into the future. For deep space objects, the 19th SDS uses an *ellipsoid screening* method in which covariance is estimated as a fixed-size ellipsoid.[59] Potential collisions are flagged if any RSO enters the volume of this ellipsoid. For near-earth objects, the propagated covariances of RSOs are used to calculate a probability of collision at the TCA.[60]

The probability of collision for RSO pairs is calculated, and probabilities that exceed a certain threshold are added to the attention list for closer observation by the SSN, or a conjunction message is sent to the O/O of the respective RSOs.

Issues

OD and uncertainty quantification are the key elements for computing a reliable P_c. The 19th SDS relies on the HAC to screen the orbital trajectories of all RSOs against all other RSOs.[61] Timely and accurate updating of this catalog relies on frequent performance of the OD process and timely identification of RSOs, and there is a need here for a more responsive OD process to support the maintenance of a more current HAC for screening efforts at the 19th SDS. The current data lag between the 18th and 19th SDSs and the current eight-hour HAC update cycle can result in the 19th SDS using hours-old RSO states when starting CA computations.

Furthermore, the HAC all-versus-all conjunction screen is a large and high-fidelity computational task that exponentially grows with the number of RSOs. An accurate prescreen that can minimize the number of candidates for subsequent high-fidelity conjunction calculations can help mitigate the exponential complexity of this task.

Potential AI/ML Application

An AI/ML-based conjunction screening application would build on previous work by using OD and neural networks to predict distance and time of closest approach. Given state vectors of two initial RSOs, a neural network can be trained to predict the distance of closest approach. Although a neural network estimate will never replace the accuracy of the CA process, it can serve as a rapid prescreening step to minimize the number of false-positive RSO pairs that make it to the high-fidelity CA step.

Chapter Findings

The SDA mission is becoming more complex, and tools that support this mission can act as force multipliers to help the 18th and 19th SDSs meet increasing demands. Processes such as checking and

[59] 18th and 19th SDSs, Combined Force Space Component Command, *Spaceflight Safety Handbook for Satellite Operators*.

[60] 18th and 19th SDSs, Combined Force Space Component Command, *Spaceflight Safety Handbook for Satellite Operators*.

[61] 18th and 19th SDSs, Combined Force Space Component Command, *Spaceflight Safety Handbook for Satellite Operators*.

verifying sensor performance, handling UCTs, and resolving broken differential corrections are currently performed manually and require significant time and effort. The growing number of mega-constellation deployments, uncorrelated tracks, lost objects, and manual differential corrections are threatening to outpace the ability of the 18th and 19th SDSs to address them. Resourcing the development of tools that run constantly in the background would be helpful in mitigating the untenable growth in object correlation cases that require manual intervention as the celestial neighborhood grows.

Noncooperative mega-constellations present unique challenges to USSPACECOM, including the deployment of secret payloads and non-Keplerian orbits that require more-frequent observations to avoid breaking ADC's thresholds. The addition of the characterization element to SDA's mission demands requires increasing person-hours to address, whereas enlisted technical expertise is constantly being lost because of tour-of-duty limitations. Launch CA processes are struggling because of the exploding tempo of launches, both cooperative and noncooperative. The changing demands in the space domain require SDA to process and respond at an operational timescale, which currently presents barriers to quicker SDA responsiveness.

To address these challenges, AI/ML tools should help shift human manpower to tasks that do not lend themselves to checklists and algorithms. Increasing sensor observation capacity could promote the ability to detect and respond to nonnominal and noncooperative behavior more rapidly. Building in slack to sensor observation capacity would help increase responsiveness to nonnominal behavior, such as maneuvering, fragmentation events, and non-Keplerian orbits. An overview of these tools is provided in the box below. The companion report offers a detailed description of the proof of concepts we developed in many of these areas,[62] and Appendix B in this report provides a brief summary of these tools for the reader's convenience.

Summary of Potential AI/ML Tools

Sensor tasking, scheduling, and prioritization	• These tools automate calibration tasking. • These tools track sensor attributes and determine tasking feasibility.
Data collection and integration	• Computer vision may help disambiguate RSOs. • These tools enable new methods of observation.
OD	• Neural networks can approximate complex orbit models in a fraction of the time.
Catalog maintenance	• Some aspects of correlating UCTs can be automated. • Generate automatic track association recommendations.
CA	• Neural networks can be a rapid prescreener for CA.

AI/ML tools have several potential impacts. These tools can improve resource allocation, optimize processes, perform tasks more quickly and frequently, and reduce the need for human analysis. These

[62] Tran et al., *Artificial Intelligence and Machine Learning for Space Domain Awareness*.

solutions act as force multipliers. According to our CA process analysis, the following impact characterizations can guide the development of AI/ML tools to meet SDA mission needs in the following areas:

- optimization of sensor tasking and prioritization to increase observation capacity and support rapid and accurate OD
- improved detection of nonnominal behavior in sensors to more rapidly identify erroneous sensor behavior that would otherwise pollute the track history of any observed satellites, thereby supporting HAC health
- increased number and frequency of sensor observations to improve the accuracy of OD, which supports arriving at more-accurate P_c predictions quickly; this can be achieved by optimizing existing sensor resources or ingesting additional data from commercial sensors
- optimization of sensor allocation, including data fusion, to take advantage of the growth in commercial sensor networks while mitigating declining sensor performance with age and the strengths and weaknesses of different sensors for different observation use cases.

By leveraging the capabilities of AI/ML, higher sensor revisit rates and faster OD can be achieved, thus acting as a force multiplier on static sensor and computational resources. In the next chapter, we will assess the quantitative impact of improved sensor allocation OD speed on the SDA mission set.

Chapter 4

Improvement Models for Characterizing Impact

Through consultation with stakeholders, we identified key areas of potential impact based on the most-acute needs for meeting the growing challenges of the SDA mission and the CA process (Chapter 2). We then assessed the role that AI/ML could play in those key areas of impact (Chapter 3). This chapter examines the nature of that impact—that is, how could improvement in one process element affect overall mission success?

Characterizing the Value of AI/ML for CA

A key area for potential improvement for AI/ML tools is in improved sensor optimization. Sensor resources are finite, and using AI/ML tools to better leverage existing resources could facilitate higher numbers of useful sensor observations and higher revisit rates. More-effective prioritization of sensor observations—i.e., to better track a high-interest object—will better direct existing sensor resources to high-priority intelligence needs.

We examine the potential cascading impacts that AI/ML tools can bring to the SDA mission by examining two key performance parameters that represent broader CA process objectives:

- **Improve the accuracy of the calculated P_c,** which avoids collisions through timely warnings.
- **Decrease probability of false alarms, P_{fa},** minimizing the false prediction that a conjunction would occur, which can cause unnecessary satellite maneuvers that degrade lifespan and mission performance.

The first improvement described below is about the impact of AI/ML tools if they can be applied to **optimize sensor tasking to increase revisit rates**, and we examine **the impact of revisit time on downstream process elements**. The second improvement described below is about the impact of **more-frequent sensor updates on the OD process**, which in turn has downstream effects on the uncertainty associated with the P_c calculation.[63]

[63] These two improvements were chosen based on current obstacles as identified by stakeholders within Space Delta 2—that is, maintaining awareness as more mega-constellations are launched, maintaining tracks when more RSOs are maneuverable, and maintaining responsiveness and accuracy in P_c calculations.

Improvement 1: AI/ML Supports More-Efficient Use of SDA Sensors, Resulting in Higher Revisit Rates

The evolving space environment is seeing increasing activity from U.S. strategic competitors, including recent behavior exhibited by China's Shijian-21 satellite. In December 2021, Shijian-21 approached a defunct BeiDou navigation satellite to conduct a sophisticated rendezvous and proximity operation (RPO), docking with the satellite, then conducting an engine burn that culminated with the Shijian-21 towing the defunct satellite to a graveyard orbit above geostationary earth orbit (GEO).[64] Although this capability has the potential to be used for responsible debris mitigation in space, there are also growing concerns about such sophisticated RPO activities enabling offensive counterspace capabilities. For this reason, the United States places a high priority on tracking and characterizing these events, and the SDA mission will be increasingly focused on these events as competition in space continues and grows. Another example of orbital maneuvering is from analysis conducted by COMSPOC of China's Shiyan-12 satellite executing complex maneuvers to avoid being observed by a U.S. satellite, the USA 270.[65]

Maintaining awareness of activities, such as those conducted by Shijian-21, requires a high revisit rate from observing sensors and full orbit coverage capability to support U.S. space operations that require SDA data in an operationally relevant time frame. Efforts to increase sensor revisit rates can make SDA operations more responsive to maneuvering satellites, and the below proposes a model for illustrating the potential impacts of increasing revisit rates when tracking maneuverable RSOs.

We present a simple analysis illustrating the potential impact of a delta-v (velocity) maneuver on the satellite position. The intent is to show the importance of the revisit rate to help maintain track on a maneuvering satellite. The example analysis considers a space object orbiting in GEO and estimates its position over a 24-hour period with and without a maneuver; the difference between the two cases illustrates the difference in the satellite position over the 24-hour period.

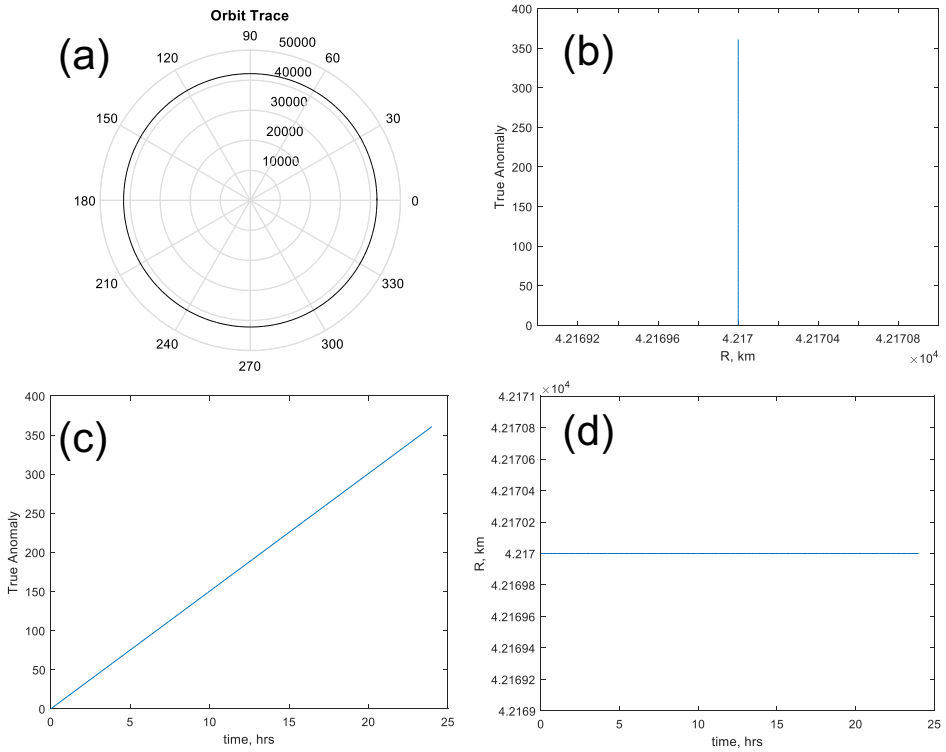
The four panels in Figure 4.1 illustrate a nominal geostationary orbit.[66] Panel (a) illustrates the trace of the circular orbit of the satellite, and panel (b) provides the radius as a function of true anomaly (or angular position). Panel (c) shows the time history of the orbit radius, which is constant, given the analysis approach, and panel (d) provides the time history of the true anomaly of the object, which has a constant slope representing a nonaccelerating body.

[64] Jones, "China's Shijian-21 Towed Dead Satellite to a High Graveyard Orbit."

[65] Werner, "An In-Orbit Game of Cat and Mouse."

[66] The analysis uses Kepler's orbit equation and does not account for any perturbations, but these approximations are sufficient for the purpose of this discussion.

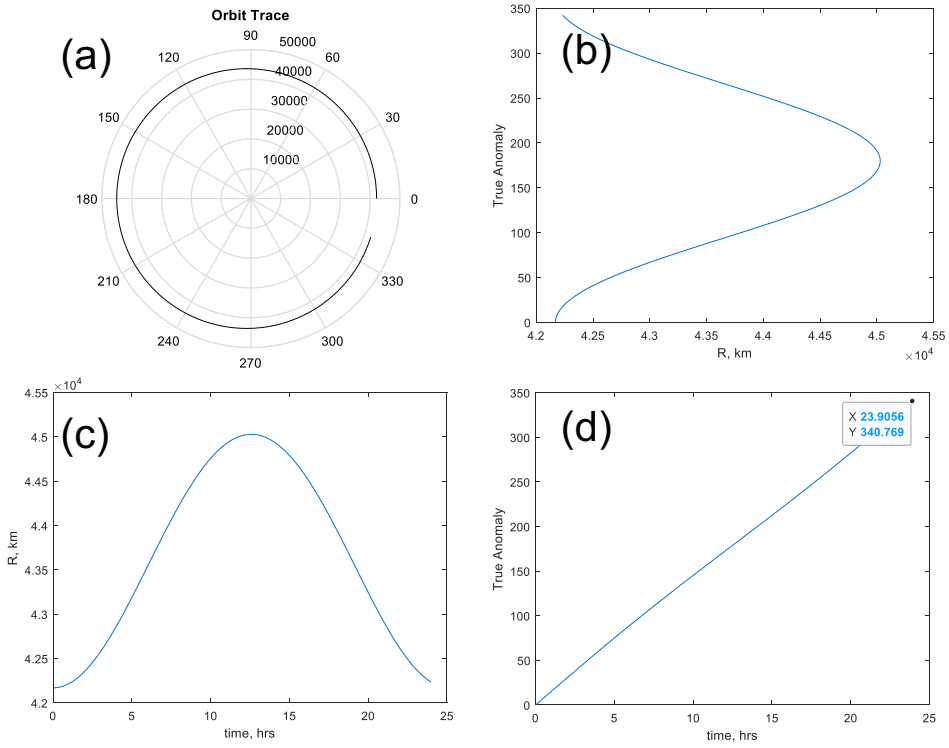
Figure 4.1. The Orbit of a GEO Satellite with No Maneuvers



NOTE: The panels depict a satellite in geostationary orbit. Panel (a) is trace of the circular orbit of a satellite. Panel (b) is the radius as a function of true anomaly. Panel (c) is the time history of the orbit radius. Panel (d) is the time history of true anomaly.

The second set of panels in Figure 4.2 illustrates the impact of a 50 m/s delta-*v*, which increases the eccentricity of the orbit to be slightly elliptical. The instantaneous thrust results in an angular drift rate of approximately 0.74 degrees/hour, or 544 km/hour in GEO—i.e., the satellite is 544 km away from its expected position after one hour. Consequently, a ground-based sensor with a 1-degree field of view—or 650 km field of view at GEO—would likely fail to acquire the satellite if it was unaware of the maneuver, since the sensor would likely attempt to place the satellite at the center of its field of view.

Figure 4.2. The Orbit of a GEO Satellite with 50 m/s Maneuver



NOTE: The panels depict a satellite following a 50 m/s delta-*v* maneuver. Panel (a) is trace of the elliptical orbit of a satellite. Panel (b) is the radius as a function of true anomaly. Panel (c) is the time history of the orbit radius. Panel (d) is the time history of true anomaly.

A search pattern would likely enable a sensor to acquire the object relatively quickly, since it is within half the sensor field of view. However, recall that sensor tasking currently occurs every 24 hours and that orbit propagation occurs once every eight hours. Supposing that we gain situational awareness of the maneuver immediately, there are no processes to dynamically retask a sensor to revisit this object. Assuming that the daily sensor tasking had an a priori revisit rate set at eight hours, the object would be about 4,000 km away from its expected position.[67] **This difference of expected versus actual satellite position of several thousand kilometers will result in a much longer search time and possibly an uncorrelated track that requires additional sensor and operator resources to resolve.**[68] The probability of the timely acquisition of a maneuvering satellite decreases as the revisit period and magnitude of the maneuver increases.

Improvement 1 illustrates that improving sensor utilization has cascading effects, such as higher revisit rate, fewer lost objects (uncorrelated tracks), and faster sensor acquisition times. All these

[67] The calculated value is the position error in three dimensions; however, the value would be less when projected on a two-dimensional sensor detector because of the relative angle between the trajectory trace and the sensor detector.

[68] The extent of the additional required resources to find the satellite after its maneuver depends on several variables, including the characterization level of the object, the actual maneuver performed, the number and type of sensors used to perform the search, and the illumination conditions, as well as the algorithms and analytics used to support the search, and cannot be easily approximated.

effects reduce analyst bandwidth at the 18th and 19th SDSs and support an increase in P_c accuracy. There are likely numerous ways to improve sensor revisit rates, and AI/ML tools may be one such approach.

Improvement 2: AI/ML for Improving Covariance Calculations in the OD Process

The OD process is foundational for maintaining HAC accuracy and identifying collision risk. In addition, OD helps identify where sensors need to take another look; in this way, improving the OD process can help prioritize which and how different SSN sensors are used.

We propose a CA process that leverages an AI/ML tool to accelerate the orbit projection estimation and prioritize the sensor scheduling based on P_c. Once the initial P_c calculation is completed, it is currently used to inform the prioritization of sensor observations. The change that we are proposing here is the use of an AI/ML tool to accelerate the subsequent OD from these sensor observations so that false alarms are more efficiently removed from the concern list. The process steps are described below and illustrated in Figure 4.3:

1. The first step (labeled 1) provides the reference orbit for all the objects being tracked.
2. Using the information from step 1, the P_c between any two objects is calculated and sorted.[69]
3. Sensor observations are prioritized based on the *N*-highest P_c and add the high-interest objects regardless of their associated P_c.[70]
4. Sensor observations are scheduled according to prioritized list.[71]
5. Sensor observations are executed.
6. The OD of all observed objects is updated. This is where the AI/ML tool is introduced to accelerate this step. We propose that an AI tool would enable this step to be performed in less than one hour to support an hourly sensor scheduling tempo.[72]
    a. Step 6a involves updating the covariance for all the observed objects. Our assumption is that information derived from the latest observations of objects cannot be used to refine the covariance of objects that were not observed—i.e., differential corrections apply only to the observed objects and cannot be leveraged for other nearby objects.
7. Position error projection is reduced for objects that have been observed.
8. P_c is updated using the latest data and calculations.

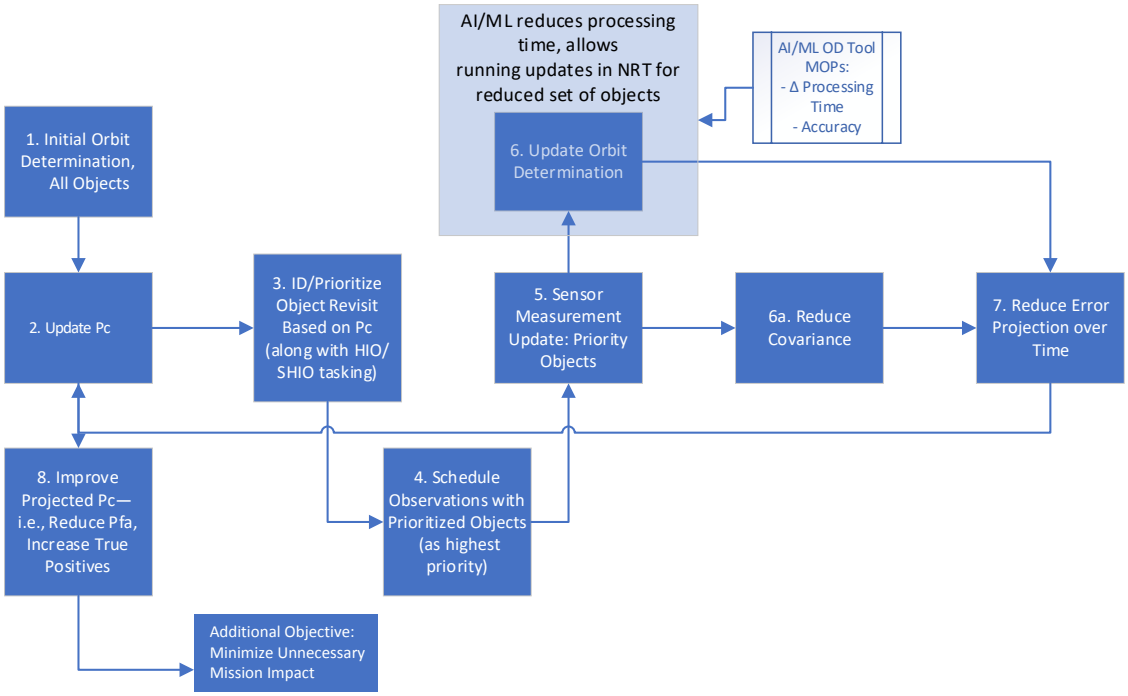
[69] For detail on the process of this calculation, see the companion report, Tran et al., *Artificial Intelligence and Machine Learning for Space Domain Awareness*.

[70] *N* needs to be determined separately based on sensor capacity and other relevant criteria.

[71] Sensor scheduling could also benefit from AI/ML and, if appropriately implemented, could improve the efficiency with which the existing sensor network is being used—that is, a possible objective function is to minimize the covariance across the cataloged RSOs. See Stottler, "Automatic, Intelligent Commercial SSA Sensor Scheduling."

[72] See Chapter 5 for a sample AI/ML-based tool to demonstrate this capability.

Figure 4.3. Overview of Proposed Modified CA Process with Dynamic Sensor Tasking



NOTE: HIO = high-interest object; MOP = measure of performance; NRT = near real time; Pfa = probability of false alarms; SHIO = super high-interest object.

Increasing the frequency of the sensor tasking cycle—e.g., from eight hours to one hour—would on its own improve the overall accuracy of P_c, assuming that the processing can also be executed within the shorter available time. However, other points in the process could be modified in parallel to keep up with this increased frequency. For example, as the tasking frequency increases, the number of objects that can be accessed and observed decreases because less time is available for the task—i.e., the sensor network is provided a shorter period to execute the observation task (SSN and possibly other supporting assets, such as commercial and allied). For this reason, an update to the sensor scheduling strategy may also be needed to support the higher tempo of sensor tasking orders. Implementation of the AI/ML tool described here also requires P_c calculations to keep up with the pace of the new sensor measurements.

Impact Characterization

We proposed a CA process in which sensor tasking is informed by both P_c and revisit rates. We use a low-fidelity model to illustratively quantify the impact of this suggestion. For this purpose, we developed a one-dimensional model consisting of 100 randomly spaced moving objects. State (speed

and position), covariance, and sensors are accounted for.[73] We can calculate covariance and P_c as a function of time and sensor measurement frequency. We set a P_c threshold of 10^{-5}, which triggers a conjunction warning.

In Figure 4.4, panel (a) illustrates the number of conjunction warnings that would be issued in a scenario in which a single sensor update is performed at $T = 0$ (no subsequent sensor updates are performed). When no measurement updates are performed, the number of conjunction warnings grows continuously with projected time. Panel (b) illustrates the number of conjunction warnings issued in a scenario in which a sensor is tasked hourly and prioritizes objects with the highest P_c. In this scenario, the number of conjunction warnings initially grows and then decreases as the sensor resolves its queue of high-P_c objects.[74] We note that all other parameters—i.e., initial object position and sensor measurement uncertainty—are the same for all the cases. Panels (c) and (d) illustrate the same scenarios for every two-hour sensor tasking and every four-hour sensor tasking, respectively. Figure 4.5 explicitly shows the relationship between the update rate and maximum number of warnings for the case described here. We see a significant reduction in the number of collision warnings when the interval between updates is reduced from four hours to two hours.

[73] We used a simple sensor error representation—measurement errors for position and speed—to form a covariance for each object. We then propagated speed uncertainty in time using a linear model to update the covariance of that object. If a sensor obtains a measurement of the object position, the position uncertainty is then reduced based on the sensor measurement. The P_c estimate is calculated next by considering the overlap between the position uncertainty of the pair of objects being considered using a one-dimensional normal probability density function to represent the position uncertainty of an object. See National Institute of Standards and Technology, "Normal Distribution," for an explanation of normal probability density functions.

The initial 100 objects are randomly distributed relative to each other and have a similar covariance based on the assumed sensor error model. The actual values of the sensor models are irrelevant for our purposes, since the effects we assessed are based on the difference between collision warnings with and without sensor updates (all other parameters are the same).

[74] We note that the baseline number of P_c represents the floor, the lowest number of objects with a P_c warning; these are objects that are sufficiently close to each other to trigger the warning, and additional sensor measurements will not eliminate them from being included in the conjunction count. The only way to decrease this baseline is to use a more accurate sensor or apply sensing techniques, such as multiple adjacent measurements, to reduce the magnitude of the objects error ellipsoid associated with the covariance matrix.

Figure 4.4. Number of Collision Warnings Assuming No Sensor Updates

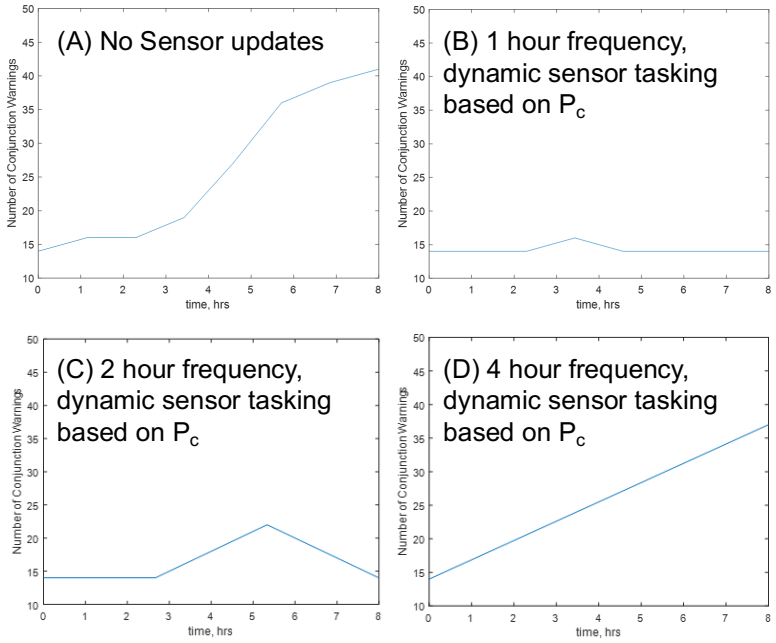
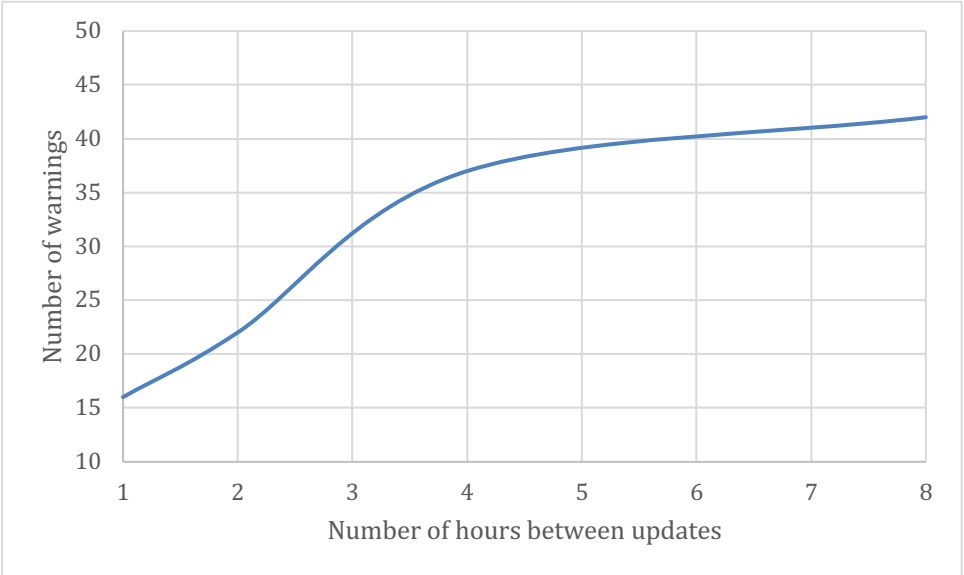


Figure 4.5. Maximum Number of Collision Warnings as a Function of Update Rate



Simple Models to Realize Impact

The model and results presented in this chapter illustrate the value of more-frequent OD updates and increased sensor tasking tempo through the use of AI/ML. Although the orbital mechanics behavior of RSOs and their relative motion are of course much more complex than the simple model

used here, the intent is to demonstrate only the concept and perform a relative assessment—that is, comparing performance with different sensor update rates—to examine the general trend that might be expected. A more quantitative assessment of the impact could be done by modeling the orbital mechanics of actual RSOs and using representative sensor models to refine the analysis presented in this chapter. We note that a similar analytical approach can be leveraged using the full catalog, along with a representation of the SSN to assess the validity and value of this strategy.

Admittedly, the implementation process requires a more in-depth examination of the next-generation systems' capabilities that operators at the 18th SDS and 19th SDSs will have (e.g., the future SDA C2 system ATLAS). Therefore, we can make only a qualitative judgment that implementing the suggested modifications, if properly performed, should not introduce any significant risk to the overall CA process, since they involve mature and well-understood processes and technologies. Further, the implementation of both these improvements simultaneously would require a balance between increasing sensor observations on more objects and more-targeted sensor retasking only on objects of high concern. In other words, implementation may require weighting the objectives of minimization of false alarms against decreasing the number of lost objects.

Chapter 5

Implementation Challenges, Conclusions, and Recommendations

The previous chapters have identified potential applications for AI/ML in the SDA mission—specifically in the CA process, in which AI/ML could help address the most-acute challenges faced by operators. Our research confirms that there is significant potential for AI/ML impact in the SDA mission. However, there are also some significant barriers that need to be addressed to enable the implementation of AI/ML tools.

Implementation Challenges

The primary barriers to implementing such AI/ML tools are based on software, hardware, and personnel. The use of legacy systems, such as SPADOC, that are written in outdated languages (e.g., Fortran) means that software is not easily scripted any more. For hardware, SPADOC is the only system that has data cables that transmit to the SSN sites, so replacing SPADOC without disrupting the SSN is a significant challenge. SPADOC has a built-in process that is based on an outdated operational tempo, but needs have morphed and now require support for emerging challenges (e.g., simultaneous reentries and nearly daily launches). Further, SPADOC cannot support the inclusion of commercial sensor data because of a lack of a hard-line connection and the difficulty of Fortran coding to add new sensors. Although ATLAS aims to address these SPADOC insufficiencies, there is some feature creep that has some analysts at the 18th SDS fearful that SPADOC might not follow through to be fully fielded.

Further, there are personnel constraints, exacerbated by the technical nature of the SDA mission. The complex mission set affects the 18th and 19th SDSs' ability to sustain a pool of uniformed subject-matter experts. The allotment of uniform billets on the floor appears insufficient to meet the rapidly accelerating growth in space activity. Training personnel is difficult, and USSF guardians change stations by the time they become proficient. Even if enlisted personnel can stay with the 18th SDS, their skill sets fetch a significantly higher compensation within the private sector. This leaves civilians (civil servants and contractors, most of whom have previously served) at the 18th SDS to serve as the enduring labor force. This enduring labor force is the only source of subject-matter experts in the world that can inform the development of AI/ML applications. However, given the uniformed personnel crunch, most civilians are pulled into operational tasks rather than having time to develop software to improve processes. AI/ML tools are not going to solve these problems, and indeed the beneficial impact of AI/ML tools for the SDA mission is largely reliant on policy, process, and technical solutions that are out of the scope of this research and heavily investigated elsewhere.

One barrier to AI/ML tool implementation is the complexity and uniqueness of the analysis required, which in turn requires a lot of work up front to define the rules for an AI/ML tool. If an AI/ML tool replaces or supplements an operator task to free up that operator, then rules that capture that operational task could be generated through observation of the human analysts, but this is difficult for highly complex tasks. Further, if the complexity of the orbital environment increases, then so does the requisite complexity of the analysis. Coupled with constraints of SSN and intentional obfuscation by adversaries, novel behavior could arise requiring human ingenuity to solve. **AI might not be well suited for some of the complex tasks required by the SDA mission, and the potential for these tools to support the SDA mission is only as good as the dataset available to develop the tool.**

Data availability also inhibits AI/ML tool development. AI/ML tools require a sufficient amount of data to capture a representative spectrum of behavior and outliers, and these data are not always there at the appropriate quantity and quality to support these tools. O/Os are not required to provide ephemerides to the 18th SDS, and while some do volunteer these data, there is an increasing number of noncooperative operators and actors in space that will not likely provide such data. Feedback from sensor observations is also not captured in a way that would support improving the optimization of tasking. Sensor sites are sensitive to comparative performance and will prefer satellites that are easier to collect on to improve their overall performance metrics. Data exist on whether an observation is collected, but there is not a systematic mechanism in place to collect feedback on whether the *right* or *sufficient* data were collected to meet the need of the request. **The development of a feedback system here could provide data to AI/ML tools looking to optimize sensor tasking effectiveness.**

Some of the challenges faced by the SDA mission are not well suited for AI/ML tools to address. For example, although AI/ML tools can help optimize how the SSN resources are leveraged, there is currently insufficient coverage in the southern hemisphere for tracking highly elliptical southern perigee orbits, and this limitation can be solved only through procuring additional sensors in that region. However, there are other areas in which there is currently insufficient sensor availability that are better candidates for impact from AI/ML. There are also sensor resolution limits and constraints because of sensor phenomenology that give an upper bound to how much more these SSN resources can be leveraged, even with AI/ML tools providing more-effective optimization of sensor allocation.

Although AI/ML tools cannot solve the lack of personnel dedicated to the AI/ML mission, these tools can be force multipliers to address this challenge, and there is a large potential for impact through supplementing or taking over human tasks. For example, there is limited manpower to support the SDA mission, and this in turn limits the availability of personnel to support tool development. Complicating this is the pace of the operations floor at the 18th SDS and 19th SDS as space becomes more contested. Support contractors are pulled into operationally focused tasks that are time sensitive rather than tool development, whereas analysts have little free time to assist with tool development or the implement of new tools into their existing workflow.

There are a lot of potential game changers with AI/ML tools, but, overall, realizing a game-changing impact requires implementing enabling technology, software, hardware, processes, or policy.

Conclusions

Realizing impact from AI/ML tools can be optimized with both tool development and implementation, as well as modifications within the SDA mission processes. Our conclusions from this research reflect both.

Tool Development and Implementation

Because of the growing demands and changing nature of the SDA mission, AI/ML tools have a high opportunity for impact as force multipliers for SDA operators. The SDA mission is increasing in scope and complexity, and, as a result, operators are challenged to address growing and changing SDA needs. Many processes, including the process for checking and verifying sensor performance, processing uncorrelated tracks, and broken differential corrections, are currently performed manually and require significant time and effort. The amount of uncorrelated tracks, lost analyst objects, and broken differential corrections is threatening to outpace the ability of the 18th and 19th SDSs to address them. Supporting AI/ML tools that run constantly is critical for mitigating the untenable growth in the proportion of object correlation cases requiring manual intervention as the celestial neighborhood grows.

Noncooperative mega-constellations will present their own unique challenges to the SDA mission. Deployments of scores of different satellites at once, without O/Os providing ephemerides to the 18th SDS, could provide cover for the deployment of secret payloads; additionally, mega-constellations in low earth orbit will presumably have to constantly maneuver as Starlink and OneWeb do, meaning that their orbits will be non-Keplerian and therefore require more-frequent observations to avoid breaking ADC's thresholds.

Any efforts to support the SDA mission need to help meet these challenges, and AI/ML tools have the most potential for impact if they can help operators perform current tasks faster and more efficiently.

AI/ML tool development that supports more-optimized sensor tasking could have a cascading impact on the rest of the SDA mission. AI/ML tools that better optimize sensor allocation, which includes the ability to integrate and fuse additional data sources, can help take advantage of the growth in commercial sensor networks while also mitigating declining sensor performance with age and the different strengths and weaknesses of different sensors for different observation use cases. This could also help increase sensor observation capacity, which could in turn enhance the ability to detect and respond to nonnominal and noncooperative behavior more rapidly. Responsiveness to anomalous behavior would be increased by building slack in sensor observation capacity.

A higher sensor revisit rate can also have cascading impacts that can help address increasing satellite maneuverability. The probability of a timely acquisition of a maneuvering satellite decreases significantly as the revisit period and the magnitude of the maneuver increase. Enabling this increased revisit rate can help better identify maneuvering RSOs, maintain HAC accuracy, and more quickly identify anomalous behavior. If revisit rates are not increased, resources required to reacquire the maneuvering satellites will continue to grow in significance as a burden on analysts. In addition, AI/ML tools could be leveraged in support of OD calculations, and impact could be magnified by

concurrently introducing other AI/ML tools in the process. Developing an AI/ML tool to optimize sensor tasking would be complementary to better OD, and the impacts from each would be magnified.

Better quantification of risk and uncertainty tolerance can support improved performance of AI/ML tools focused on prediction and classification. Our proof-of-concept example for CA illustrates a trade-off between the number of false positives identified and the confidence that all potential collisions have been identified. Missing a potential collision is likely more consequential than flagging a possible collision that turns out to be a false positive, but the consequences of too many false positives are increased process and analyst time required to reevaluate the potential collision and potentially increased sensing resources if the 18th SDS needs to retask.

We find that the level of uncertainty and risk an SDA operator is willing to accept has consequences for the performance of the model. When used in collaboration with an SDA operator, the model prediction of the closest approach and estimate of the prediction uncertainty provide a way to rapidly screen through a large number of combination pairs, but the performance of this tool is dependent on the risk the SDA operator is willing to accept. Better classification performance can be achieved when uncertainty quantification is considered in collaboration with an SDA operator.

There are trade-offs associated with data used to train AI/ML models as well. AI/ML tools require high-quality training data to both improve performance and expand the initial state parameter space. Anyone looking to use SDA training data should be cognizant that there exist no "truth" data. All inputs into AI/ML algorithms have inherent uncertainties because of such factors as measurement error and numerical precision. These uncertainties in turn have implications for how to optimize AI/ML performance and benchmark against current algorithms. Although this is especially true for synthetically generated training data, this will also be the case with data pulled directly from the CAVENet or SPADOC systems.

SDA Process Changes

AI/ML tools cannot address all the challenges of the SDA mission, but process changes can drive greater impact from these tools. As SDA takes on the role of support to a warfighting domain, SDA processes need to be able to support decisionmaking at an operational time frame. This approach requires a process that is more responsive to anomalies, that can more quickly ingest and analyze sensor data, and that has sufficient capacity to reallocate sensor and analyst resources. Currently, there are limitations in the timeline that present barriers to quicker SDA responsiveness. AI/ML tools can help address these limitations, but these limitations need to be looked at holistically to maximize potential impact.

For example, the limiting factor in the CA process is the time to propagate the orbits at the 18th SDS, send the data over to the 19th SDS, and conduct screening on all pairs. Process changes that can accelerate the exchange of data between the 18th SDS and 19th SDS, which currently takes two hours, could facilitate a more responsive screening process, which in turn produces more-reliable information for sensor retasking. Until the timing limitations inherent in the current CA process are overcome, there is an upper bound on the impact from other AI/ML tools that support better sensor resource optimization and quicker OD.

Realizing significant impacts from AI/ML requires moving to an architecture that enables more AI/ML development and fielding. There are numerous efforts to build and implement AI/ML tools for the SDA mission. The significance of impact from AI/ML tools is dependent on addressing architecture constraints. However, planned architecture changes will affect the significance and feasibility of realizing the impact from these AI/ML tools. For example, continued reliance on SPADOC and CAVENet means that realizing the benefit of improving sensing technologies on orbit is limited by data processing in these legacy systems. Implementing new tools in the SDA mission also requires the ability to test and validate these tools, and an important element of these architecture changes will be enhancing the ability to test and validate new AI/ML tools. Architecture changes may enable higher impact from AI/ML tool implementation, but AI/ML tool design should be optimized for the architecture of the future to maximize this impact.

Recommendations

SSC SDA Acquisition Delta (SSC/SZG) and the Air Force Research Laboratory should provide clear guidance to AI/ML tool developers to focus on tools that address the needs of operators today but can continue to provide benefit in a future SDA architecture. To realize the impact of AI/ML tools on the SDA mission, resources to update the SDA architecture should be prioritized, as these modifications are a key enabler to AI/ML impact. There is a substantial level of effort that has been ongoing for decades to modernize the SDA architecture, but the challenges are significant. Addressing these challenges requires continued effort and prioritized allocation of resources to these architecture updates and will in turn build the backbone of AI/ML implementation into the SDA mission. However, a recommendation to addressing aging SDA architecture is not a new one, nor particularly helpful to the myriad stakeholders that have been seeking to accomplish this. Although the recommendation is important, our project team instead was focused on characterizing impacts. Since these impacts will vary based on potential architecture changes, we recommend a focus on investing in AI/ML tools that either *enable* a transition to a modernized architecture or are designed to *leverage* this modernized architecture. Engagement with stakeholders working on Kobayashi Maru, for example, can help shape AI/ML tool development, and tool developers should ensure that they are working in a realistic environment for *future* deployment.

AI/ML tool development should focus on force multipliers that help meet the growing detection and characterization challenges faced by the 18th SDS and the 19th SDS. SSC/SZG should look for opportunities to invest in these tools. Improvements here can provide a foundation for other AI/ML tools and impacts that could support the SDA mission in more-nuanced and even unanticipated ways, but this foundation first needs to be reinforced for the growing and evolving SDA mission.

SDA operators, in particular at the 18th SDS, 19th SDS, and National Space Defense Center, should better articulate mission needs, sensor requirements, and acceptable uncertainty to optimize potential enhancements from AI/ML tools. Since AI/ML tools are only as good as the rules they are given, more clarification is needed about what these rules should be, on the basis of criteria for mission effectiveness. That clarification requires a detailed look at what these metrics should be moving forward. Confidence measures and acceptable risk can help improve the

characterization output of AI/ML tools. Sensor tasking tools can be better supported through more-explicit data requirements that capture when and whether an observation is sufficient according to RSO prioritization and intelligence needs. Efforts are ongoing at the 18th SDS to change the way mission tasking orders are articulated, including specifying intent, priority, and what the observation is seeking to achieve to better task sensors. Continued support for these efforts will enable higher impact from AI/ML tools designed to support sensor tasking and prioritization.

USSF Space Operations Command, via the SDA Mission Area Team, should examine where SDA processes could be modified to enable more-significant impact from AI/ML tools. Realizing the impact of AI/ML tools is in many ways not about the technology itself but about process improvements to enable their capabilities. These improvements include identifying the metrics for mission effectiveness, assessing existing processes through feedback mechanisms, and addressing manpower bottlenecks. By improving these processes, AI/ML tools can be trained to optimize sensor tasking and prioritization, leading to improvements in efficiency, accuracy, and responsiveness.

AI/ML tool development should focus on those tools that enable more-effective and more-efficient sensor tasking. SSC/SZG should seek ways to acquire and develop those tools that can be implemented to support better sensor tasking. Efforts at the 18th SDS to capture observation intent should also be supported, as these efforts in combination are key enablers, and this process element presents a clear point for impact across the SDA mission. Better utilization of the existing SSN will in turn facilitate better OD, more-responsive CA, and improved SDA overall.

SSC/SZG should continue to support efforts in AI/ML tool development and ensure that there are processes and infrastructure in place to test and validate these models. The ability to build trust in AI/ML models is vital to the success and operator adoption of these tools. Ensuring the availability of sandboxes for developers to leverage to validate models is essential to building the trust needed to field new AI/ML tools. Access to these sandboxes and shared datasets should be supported, and bringing new tools into these sandboxes should be a focus of USSF procurement efforts.

The development of AI/ML tools requires significant investment in ensuring high-quality training data, and investment in tools should include a consideration of this availability. The availability of high-quality training data is another enabler for AI/ML impact, and the USSF should support efforts at SSC in its Cross Mission Data team to ensure the availability of these data to AI/ML tool developers. AI/ML can be powerful at learning complex relationships between inputs and outputs of a task, and AI/ML is essentially a universal function approximator that learns from example. To develop effective AI/ML applications for SDA tasks, then, it is essential to start collecting input-output data from analysis tasks to curate a database that can train future systems. This enables AI/ML to learn from best analyst practices and slowly automate more-complex tasks as the database grows. Given the evolving nature of the space environment, this effort will need to be continuous.

Appendix A

Overview of Space Domain Awareness Functions

SSA is a subset of SDA that focuses on the orbital segment. The SDA functions outlined in the January 2022 space doctrine note from the USSF provide an overview of SDA and the orbital segment, SSA. The 2020 Joint Publication 3-14 provides a more detailed view of SSA functions and describes SSA overall as the requisite foundational, current, and predictive knowledge and characterization of space objects and the operational environment on which space operations depend.[75] These functions are each described below to provide the reader with an overview of SDA activities and objectives.

D/T/ID

The D/T/ID function is observation based and includes the ability to "search, discover, and track spacecraft and events; distinguish objects from others; and recognize objects as belonging to certain types of missions."[76] The function supports collision avoidance by considering current and future position of RSOs and provides a basis for object characterization.[77]

Characterization

The characterization function captures the features and employment of an RSO. Characterization involves integrating information from terrestrial and orbital sensors to support space operations and assess threats.[78]

Threat Warning and Assessment

Threat warning and assessment evaluates potential threats to space operations and their impact on an RSO's mission and ability to deliver effects.[79]

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[75] Joint Publication 3-14, *Space Operations*, p. ix.

[76] Joint Publication 3-14, *Space Operations*, p. 22.

[77] USSF, *Operations*.

[78] USSF, *Operations*.

[79] USSF, *Operations*.

Data Exploitation and Integration

Data exploitation and integration include the exploitation of multiple data sources and the fusion of these data to support decisionmaking. Successful data exploitation and integration provide higher confidence and earlier predictions.[80] Improving these metrics as space becomes more congested and contested requires increased speed and improved data quality.

[80] USSF, *Operations*.

Appendix B

Summary of Artificial Intelligence and Machine Learning Tool Development and Proof of Concept

This report discusses how AI/ML tools might affect SDA processes through infrastructure changes, the addressing of operator needs, and relationships among various elements of the SDA process. Our research identified implementation barriers and limitations in tool development, such as data availability, accuracy trade-offs, and inherent risks and uncertainties. This information can guide AI/ML tool development and investment by identifying areas of impact and avoiding known pitfalls. This appendix provides an overview of our proof-of-concept development—including methods, observations, and lessons learned—to assess the potential impact of AI/ML tools. **For a more comprehensive discussion of methodology and results aimed at an SDA operator or an AI practitioner audience, please refer to the companion report.**[81]

In this appendix, we summarize two BNN case studies we designed to better understand the application of AI for the SDA mission. Both are stand-alone applications: They may assist operators in limited aspects of SDA, but, cooperatively, the two case studies may alleviate the computational and manpower demands of CA analysis. BNNs were chosen for their ability to model probabilities and output confidence and uncertainty. Quantifying uncertainty is an important aspect of gaining operator trust and encouraging use of these tools.

These use cases are selected for two reasons: (1) the availability of training data representative of what are used by SDA operators and (2) a desire to deconflict with use cases being studied at other organizations (e.g., sensor orchestration as part of the MACHINA effort). The two case studies are built with consideration of the current SDA processes in mind and with buy-in from operators from the 18th SDS. Although the case studies might have applicability in future SDA data infrastructure architectures, many of the design considerations would need to be reevaluated.

Case Study 1: Elliptical Screening Tool

The current SDA conjunction screening process uses two algorithms: elliptical screening (flag if two RSO states are projected to come within 25 km of each other) and probability of collision threshold (propagate RSO covariance to TCA). We developed a BNN to approximate the elliptical screening tool. Specifically, the BNN is provided with two RSO states at a given time and produces a

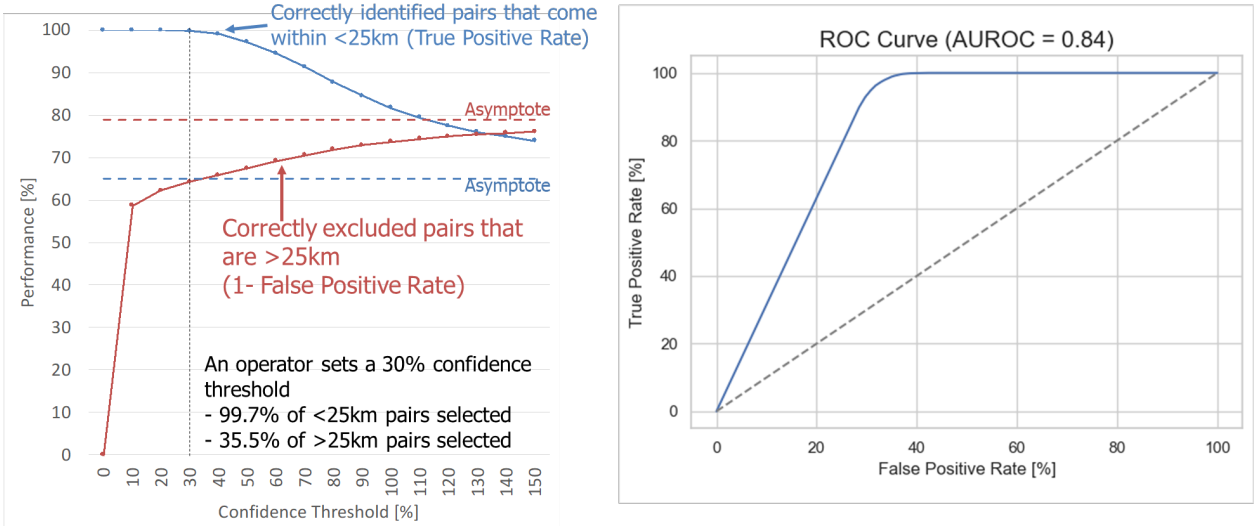
[81] Tran et al., *Artificial Intelligence and Machine Learning for Space Domain Awareness*.

prediction of the distance of closest approach over the next five days, as well as an estimate of uncertainty (covariance) in the prediction. These outputs inform a postprocessing classifier to flag a pair of RSOs for a possible conjunction event. If flagged, additional resources and more-accurate algorithms can be dedicated to better characterize the possible conjunction. In the analysis presented here, we constrained the synthetic data to Starlink constellation satellites for a proof-of-concept example.[82]

We assessed performance by calculating how many RSO pairs are correctly flagged. We virtually divided the RSO pairs into two groups: Group A includes data points with a true closest approach less than or equal to 25 km, and group B includes data points with a true closest approach greater than 25 km. The goal is to flag as many objects as possible within group A while minimizing group B flags.

Using our synthetic data, we fed in all possible RSO pairs to the BNN. The postprocessing algorithm analyzed each BNN output and flagged a pair if the predicted distance of closet approach was less than or equal to 25 km or the prediction uncertainty exceeded a threshold. This threshold can be set by an SDA operator to represent the level of uncertainty and risk they are comfortable with. In Figure B.1, we present the performance trade-off for group A and group B as a function of this threshold.[83]

Figure B.1. Performance Optimization



NOTE: ROC = receiver operating curve; AUROC = area under ROC.

[82] To generate ground truth data, we ran Monte Carlo simulations of Poliastro orbit models, a Python astrodynamics package (Poliastro—Astrodynamics in Python, homepage). Additional historical data available to the 18th SDS and the 19th SDS likely could be leveraged to expand the data for the initial orbital state parameter space to further refine this analysis.

[83] We define the *confidence metric* as the percentage of standard deviation with respect to the mean prediction. A confidence metric of zero indicates that the SDA operator is skeptical of all BNN predictions, and thus every combination pair is flagged for further review, resulting in no added value from the BNN. On the opposite end of the spectrum, as the confidence metric approaches infinity, this is equivalent to an SDA operator having absolute trust in the BNN prediction. Doing so results in a significant number of misclassifications for group A.

For example, if an SDA operator sets a 30 percent confidence threshold, the blue curve (true positive rate) shows that the algorithm will correctly flag 99.7 percent of RSO pairs in Group A. This threshold also excludes 64.5 percent of elements from Group B, which results in a reduction of overall RSO pairs that will require higher fidelity calculations in subsequent CA steps. In collaboration with an SDA operator, the inclusion of uncertainty quantification may improve the performance of prediction and classification methods.

It is noteworthy to mention the computational efficiency inherent to this model because the BNN does not require an evaluation of the orbit propagation trajectory. When extrapolating computation time to a realistic number of objects, the BNN system takes approximately 77 seconds to screen 1 billion RSO pairs for possible conjunctions.[84] When compared with the process time frames shown in Chapter 2, the BNN system presents an opportunity to reduce the computational load, increase flexibility, and increase responsiveness by identifying and flagging likely conjunction events without requiring reference to the HAC trajectories.

Case Study 2: Orbital State Propagator Tool (Description, Metrics, Lessons Learned, and Outcomes)

Many SDA processes requires predicting the future state (position) and covariance (uncertainty cloud) of an RSO. Examples include (1) knowing where to point ground station sensors for observations and (2) the calculation of probability of collision. We tested the viability of a BNN-based tool to propagate both state and covariance. Such a tool gives SDA operators the ability to quickly evaluate future RSO trajectories with minimal computational load and without accessing the trajectories stored on CAVENet. This tool can create flexible timelines for covariance-related analysis not amenable to current processes, such as dynamic sensor retasking.

The BNN ingests the state and covariance of an RSO at a given time and outputs the state and covariance at a specified time in the future.[85] For this prototype, we trained the BNN to output an RSO's covariance at a fixed future time. We used various statistical measures to measure performance to assess how closely the BNN uncertainty matches the true covariance (as informed by Monte Carlo simulations of orbit propagation).[86]

In general, the BNN was accurate at predicting the future RSO state but overestimated the standard deviation. Figure B.2 illustrates the state accuracy (via MAPE) on the left and the covariance

[84] Recent estimates state the 18th SDS maintains a catalog tracking more than 44,000 RSO (Space-Track.org, homepage). The number of unique combination pairs is then 967,978,000. The algorithm performance is evaluated on a NVIDIA T4 Tensor Core GPU.

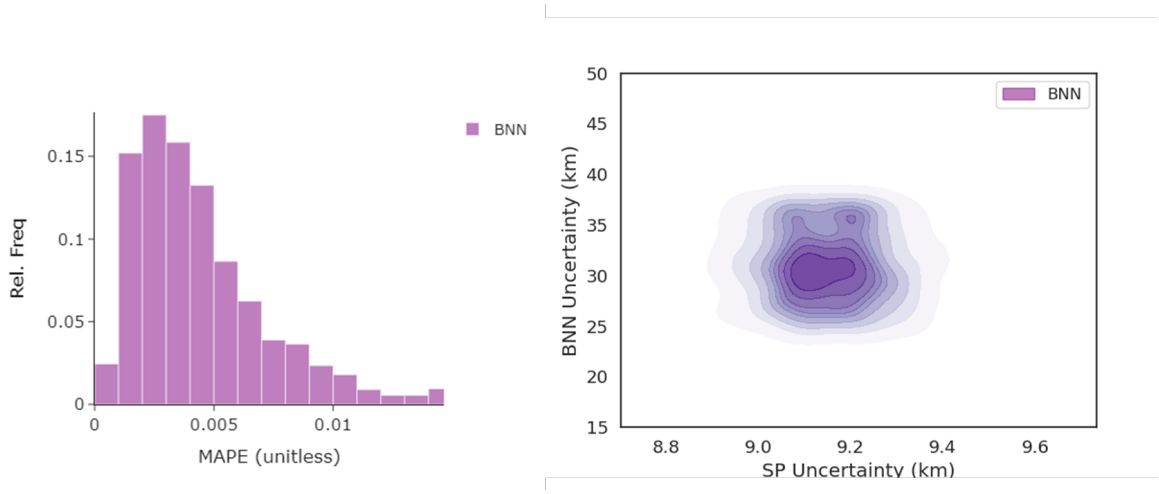
[85] Using a RAND in-house special perturbation model, we propagated a series of initial-state probability distributions forward in time, eventually creating a training dataset consisting of ~200,000 row of events. The initial state distributions were sampled from Starlink constellation orbits.

[86] We evaluated the performance of these models across three separate metrics, which are commonly used to assess differences between probability distributions: MAPE, covariance RMSE, and Kullback-Leibler divergence. Positional MAPE is the percentage error between the final-state coordinates calculated from the special perturbation algorithm and our model predictions. Covariance RMSE between the covariances is obtained from our special perturbation calculation and our model. Kullback-Leibler divergence is a metric that incorporates differences in both covariances and means between the distributions generated by our model and our special perturbation calculation.

accuracy on the right (via a kernel density estimation comparing x-axis ground truth against y-axis predictions).

It is challenging to contextualize these results against operational performance. We are unaware of existing research that has characterized special perturbation model performance. Special perturbation methods are currently the only algorithm available to SDA operators for propagating states and covariances in the HAC. The performance of special perturbation methods is interdependent on three key components: (1) the relative or absolute error tolerances used, (2) the physical assumptions made by the algorithm, and (3) the computing hardware capabilities at these facilities.

Figure B.2. BNN Performance at Predicting T+1,000 [s]



NOTE: In this figure, we present the aggregate performance metrics for the BNN. On the left, we show a histogram of the MAPE to demonstrate the accuracy of the BNN at estimating the mean (lower is better). On the right, we show a two-dimensional kernel density estimation of the standard deviation estimates (y-axis) relative to ground truth (x-axis). This second plot demonstrates that the BNN tends to overestimate uncertainties by a factor of approximately two or three relative to special perturbation calculations.

A BNN-based orbit propagation tool still requires additional research to determine feasibility for the SDA mission. By scaling up timings from the test set, we estimated that it takes about 12 minutes for the BNN to propagate 50,000 RSOs into the future (T+5 days).[87] For context, ASP currently takes 40 minutes to propagate the entire space catalog during each eight-hour cycle. Although the BNN is faster than current processes, we lack performance comparisons. We suggest a subsequent study that establishes current operational requirements for the SDA mission (including orbit propagation) to guide tool development. Without these requirements, it is difficult to contextualize the results in Figure B.2 and hard to determine whether the BNN is a good-enough approximator.

[87] In our computing environment, the BNN takes 10 s to propagate 50,000 RSOs through a single time step of 1,000 s into the future. Extrapolating, recursively repeating this process to propagate the entire object set through a five-day time horizon would take an estimated ~750 s.

Case Study Finding and Recommendations

In this appendix, we summarized two case studies that use BNNs to support analysis conducted by SDA operators. Both case studies attempt to use the uncertainty quantification provided by BNN in different ways that are beneficial to the SDA operator.

In our first case study, we developed an elliptical-screening-tool workflow to flag RSO combination pairs that are expected to violate the elliptical screening process. The BNN algorithm developed produces predictions of the closest approach and estimates of the prediction uncertainty, providing a way to rapidly screen through a large number of combination pairs. In collaboration with an SDA operator, the inclusion of uncertainty quantification improves the performance of the prediction and classification method. In addition, the performance of the BNN is dependent on the level of uncertainty and risk that the SDA operator is willing to accept.

In our second case study, we attempted to predict the future states and covariance of an RSO over a variable time of flight. We explored the use of two algorithms and evaluated the performance of those algorithms with different metrics. We found that different algorithms perform better or worse depending on the specific metric used, and algorithms are often susceptible to underperforming when limited in data quantity and quality. In the absence of an optimal model, the algorithm design should consider the balance of different performance metrics and implications for the quality of predictions.

Abbreviations

ADC	Automatic Differential Correction
AI	artificial intelligence
ASP	Automatic Special Perturbations Ephemeris Generation
ASW	Astrodynamics Support Workstation
ATLAS	Advanced Tracking and Launch Analysis System
AutoCA	Automatic Conjunction Assessment
BNN	Bayesian neural network
C2	command and control
CA	conjunction assessment
CAVENet	Command, Analysis, Verification, and Ephemeris Network
COM	composite
DoD	U.S. Department of Defense
DRAGON	Defense Readiness Agile Gaming Online Network
D/T/ID	detect/track/identify
GEO	geostationary earth orbit
GSSAP	Geosynchronous Space Situational Awareness Program
HAC	High Accuracy Catalog
MACHINA	mission-driven, autonomous, collaborative, heterogeneous, intelligent, network architecture
MAPE	mean average percentage error
ML	machine learning
NDPP	Non-Traditional Data Pre-Processer
OD	orbit determination
O/O	owner-operator
P_c	probability of collision
RMSE	root mean squared error
RSO	resident space object
SACT	Sprint Advanced Concepts Training
SDA	space domain awareness
SDS	space defense squadron
SP	special perturbation
SPADOC	Space Defense Operations Center
SSA	space situational awareness
SSC	Space Systems Command

SSC/SZG	Space Systems Command Space Domain Awareness Acquisition Delta
SSN	Space Surveillance Network
SuperCOMBO	Super Computation of Miss Between Orbits
TCA	time of closest approach
UCT	uncorrelated track
UDL	Unified Data Library
USSF	U.S. Space Force
USSPACECOM	U.S. Space Command

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PROJECT AIR FORCE

To address the growing demands of operating in the space domain, space domain awareness (SDA) operators must determine how to prioritize sensor observations more effectively, scale up to meet the sheer volume of resident space objects, and develop analytic capabilities that reflect the complexity of orbital mechanics and space operations, all while maintaining the responsiveness necessitated by operations in a warfighting domain. These factors present significant challenges to those tasked with the SDA mission and point to this mission as a prime candidate for support from artificial intelligence (AI) and machine learning (ML) tools, because such tools have the potential to increase the analysis tempo, expand the amount of usable data for this analysis, and free up operator time for more-complex tasks.

This report characterizes the nature of the impact that AI/ML tools could bring to the U.S. Space Force's SDA mission, with a focus on the conjunction assessment process to quantify the risk of collision in space. The impact of AI/ML tools has not been well understood, and this lack of understanding is a barrier to planning and optimizing the tools' integration. To support this assessment of AI/ML tools, the authors interviewed stakeholders, reviewed existing academic and doctrinal literature, developed detailed process maps, and built exploratory AI/ML models.

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